

NEURAL BELLMAN OPERATORS

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We develop Neural Bellman Operators for continuous-time economic models with recursive utility, preference adjustment, and strategic interaction. A critic evaluates continuation values and a feasible actor uses their derivatives to improve decisions. For a nonlinear capital economy, an exact finite Bellman identity connects occupation-weighted continuation error, action quality, and continuous-time payoff bounds. Complete sixteen-stream comparisons in ten and fifty dimensions establish economically material gains over the declared neural HJB implementation. In fifty dimensions, NBO attains the prescribed economic target with less measured work than direct policy optimization. Cached raw-costate improvement remains cheaper, and direct NBO–Raw and NBO–DPO payoff differences remain unresolved. The candidate-occupation mechanism account is numerically instantiated but does not certify positive mechanism gains. Independent scalar approximation and finite-observation experiments provide separate accuracy and implementation assessments. The original applications retain their distinct utility domains and equilibrium conditions.

KEYWORDS: Dynamic programming, Neural approximation, Recursive utility, Endogenous preferences, Viscosity solutions, Dynamic games.

1. INTRODUCTION

Many economic decisions depend on continuation values that must themselves be computed. A household chooses consumption and portfolio risk while investing in its preferences. Recursive utility makes future utility part of the evaluation equation. A sophisticated consumer anticipates the choices of future selves, and a firm invests against the policies of its rivals. In each case, a proposed change in behavior is useful only if the continuation object is evaluated accurately enough for the relevant

economic decision. The preference-formation problem of Qi (2025) brings these internal and external choices together.

This paper studies Neural Bellman Operators (NBO), a numerical evaluation and improvement map for these problems. A critic evaluates the continuation value of a specified policy. A feasible actor changes the actions entering the economic generator, holding that value and its derivative channels fixed. An independent error account evaluates the resulting policy. The numerical object includes its boundary conditions, utility domain, approximation scheme, information structure, and deployment rule. These restrictions are part of the method: they determine what the computed policy and its reported accuracy mean.

The economic role of the critic is the central question. Learning a continuation value can pool information across states and actions, but the fitting cost and approximation bias may exceed the benefit of that reuse. A positive policy gain does not establish that its critic was useful. Nor does an accurate derivative at training states establish an improvement along the distribution generated by the deployed candidate. We connect evaluation error to feasible action quality and then to economic payoffs, keeping the distribution, the evaluation target, and the numerical transfer explicit. The computational comparisons examine the same chain through direct alternatives to the learned evaluation block.

The analysis has three parts. First, the NBO specification separates policy evaluation from action improvement and states the boundary and utility conditions needed by both. Differential and positive-weight Bellman formulations give distinct numerical realizations of this map. Their error accounts distinguish policy-evaluation defects from incomplete action maximization; the monotone formulation supplies a viscosity-selection argument under its stated approximation hypotheses. These results concern the output errors of a numerical method, rather than convergence of a particular neural optimizer.

Second, a nonlinear capital economy permits a detailed connection between the computed controller and the original continuous-time problem. Log felicity, bounded consumption shares, additive diffusion, and bounded coupled production yield an analytical comparison policy and explicit propagation bounds. A policy-specific payoff identity, diffusion transfer, and simultaneous common-path

inference measure the gain of a fitted candidate and its difference from another candidate. The absolute comparison with the optimum retains its own anchor gap. An exact finite Bellman identity places continuation error along the feasible change from the reference action to the candidate action, integrated under the candidate’s occupation law. A computable maximization-gap bound records unfinished actor optimization even when global concavity of the learned continuation is unavailable. The resulting gain bound transfers payoffs to the diffusion without equating discrete and continuous costates.

Third, the computational comparison measures economic performance and the work used to attain a stated target. It executes all sixteen members of a declared finite distribution of complete algorithm streams for NBO, cached antithetic Raw improvement, direct policy optimization, and direct neural HJB fitting in dimensions ten and fifty. The experiment uses the same nonlinear capital model at higher volatility, with fixed architectures, training schedules, and stopping rules. All 128 executions return fitted policies. NBO’s schedule-relative method lower gains exceed 0.000943 and 0.000957 in the two dimensions.

The direct payoff comparison establishes a material advantage over the specified neural HJB procedure. An independently proved common-innovation transfer sharpens the deterministic allowance on the original observations, retaining the original confidence family, training, and stopping decisions. The resulting NBO-minus-HJB intervals are contained in $[0.000179, 0.000411]$ and $[0.001347, 0.001650]$; both exceed the prespecified 10^{-4} economic margin. The original intervals are reported alongside this refinement. The four NBO–Raw/DPO contrasts remain unresolved, rather than establishing equality. In dimension fifty, every NBO, Raw, and DPO stream attains the online gain target 0.0005. Their mean complete process work is approximately 357.80, 346.22, and 596.94 seconds; HJB attains the target in no stream. NBO therefore improves on this direct residual-fitting implementation and uses less measured work than DPO for the declared target, while Raw remains cheaper. These findings concern the executed finite method distributions, not all possible training designs.

The mechanism assessment separately computes every term of the finite Bellman account on the returned NBO policies. It uses the candidate’s occupation

law and independent action-bridge continuation banks, including the training-to-
assessment grid discrepancy. Its protected lower gains are approximately -0.006778
and -0.004800 , so this sufficient account does not certify a positive mechanism gain
at the executed precision. The critic-risk comparison with Raw is also inconclusive
in both dimensions. This distinction keeps the method-payoff finding separate
from a claim that costate regression has established a general risk reduction. The
earlier capital comparisons, including their unfavorable raw-costate diagnostics
and inconclusive direct payoff intervals, remain visible.

Absolute economic accuracy has its own account. The new NBO full-adapted-
class regret upper bounds are at most 0.025971 and 0.036843 , which do not certify
proximity to the optimum at the 10^{-4} margin. An independent scalar Howard
calculation instead supplies finite-grid residual and action-restriction bounds, while
its deployed classical policies outperform the frozen historical neural policies on
the measured scalar comparisons. The scalar NBO–DPO ranking reverses at one
initial state. These observations identify the scope of the approximation evidence
without changing the economic problem or treating a one-dimensional grid as a
high-dimensional reference.

The information structure also has a concrete economic implication. A manager
can observe capital at finite dates, apply a saved feasible consumption rule, and
hold that choice until the next observation. A separate experiment verifies this
measured-state implementation against its corresponding protected parent. All
four parent comparisons have positive lower gains, and six of twelve measurement
specifications retain a positive continuous-economy lower gain after their sensing
allowances. At 4,096 decision dates with exact measurements, the lower gains are
approximately 0.001051 and 0.001054 in dimensions 10 and 50. Measurement error
can exhaust this margin. The result identifies the precision required by the specified
controller rather than assuming that finite observations reveal latent innovations.

The original applications remain essential to the economic scope of NBO. Recur-
sive utility changes the fixed-policy evaluation problem. Endogenous preference
adjustment links an internal shadow value to consumption and portfolio demand.
Temporal selves require a spike-deviation equilibrium against two continuation
values, while strategic firms require a complete unilateral best response against

fixed rivals. These applications share the evaluation–improvement structure but have different economic error criteria. Their full models, derivations, proofs, and numerical records are retained in the current appendices and the indexed reproduction record.

1.1. *Relation to the literature*

Neural residual methods build on numerical dynamic programming and function approximation, not on a new principle of optimality. Projection and related methods exploit economic structure to approximate a Bellman equation; see [Judd \(1998\)](#). Derivative approximation by smooth neural networks provides useful representation results ([Hornik et al., 1990](#)), but does not establish that a particular finite architecture or optimizer solves a nonlinear boundary-value problem.

The differential critic is closely related to the neural PDE approach in [Sirignano and Spiliopoulos \(2018\)](#). Their DGM formulation includes differential and boundary conditions and treats HJB equations. The deep BSDE method of [Han et al. \(2018\)](#) also includes an HJB application. The distinguishing object here is the combination of an explicit feasible actor, block-specific evaluation, and economic error accounting across control and equilibrium problems. An analytical formula for the maximizing action is helpful when available, but is not a universal dividing line between neural PDE methods. Likewise, recursive dependence on the value is a nonlinear zeroth-order term; full nonlinearity in the PDE sense concerns the highest derivatives.

Recursive utility separates intertemporal substitution from risk aversion ([Epstein and Zin, 1989](#), [Duffie and Epstein, 1992](#)). Its infinite-horizon formulation requires particular care about utility-process selection and transversality; [Herdegen et al. \(2021\)](#) explains why a stationary algebraic solution alone is insufficient. The time-inconsistency extension is an equilibrium problem in the sense of [Björk et al. \(2016\)](#), not an ordinary control problem with a relabelled flow payoff. Finally, stochastic approximation results require their own step-size, stability, and limiting-dynamics conditions ([Borkar, 2024](#)). They do not convert constant-step Adam into a convergence theorem for economic optimality.

The article proceeds from the economic problem to its evaluation and improvement map. Section 4 states the general verification results. Sections 5–6.3 specialize the welfare account to the capital economy. Section 7 states the other economic applications. The computational study distinguishes the existing comparison from the current experiment. The supplement collects the complete models, derivations, proofs, and supporting diagnostics by subject.

2. THE CONTROLLED ECONOMY

Let W be an m -dimensional Brownian motion on a filtered probability space satisfying the usual conditions. The state $S = (u, X)$ contains internal and external state variables. An action $a = (\theta, c, \pi, \dots)$ combines internal adjustment and external choices. The controlled state evolves as

$$dS_s = b(s, S_s, a_s) ds + \sigma(s, S_s, a_s) dW_s, \quad a_s \in \mathcal{A}(s, S_s). \quad (1)$$

The compact feasible correspondence $\mathcal{A}$ is part of the model. Actions are progressively measurable and must generate a well-defined state process satisfying the declared state restrictions. State viability does not follow from bounded actor outputs alone.

We work initially on a finite horizon T . A computational domain D is either invariant, equipped with a specified reflection mechanism, or stopped on exit. In the stopped case let $\tau = T \wedge \inf\{s \geq t : S_s \notin D\}$. The payoff $g(\tau, S_\tau)$ is specified on the entire parabolic stopping boundary, including the terminal face. For a fixed admissible policy a , utility is the policy-specific solution

$$Y_s^a = g(\tau, S_\tau) + \int_s^\tau f(v, S_v, a_v, Y_v^a) dv - \int_s^\tau Z_v^a dW_v. \quad (2)$$

Adjustment costs are included in f . For example, time-additive utility has $f(s, S, a, y) = r_0(s, S, a) - \rho y$, where r_0 includes all current resource and adjustment costs. Define $V^a(t, x) = Y_t^{a, t, x}$ and $V^*(t, x) = \sup_a V^a(t, x)$. The value inside a candidate policy's recursive objective is its own utility process, not the unknown optimum inserted into every candidate objective.

ASSUMPTION 1—Finite-horizon comparison class: *The controlled state problem is well posed for the admissible policies used below. The coefficients are continuous, locally Lipschitz in the state uniformly over actions, and bounded on the compact stopped domain. The payoff is continuous and bounded. On an invariant utility interval containing all policy values, f is continuous in its arguments and uniformly Lipschitz in utility with constant $L \geq 0$. The admissible class is stable under concatenation and has the measurable-selection properties needed for dynamic programming. The resulting HJB problem and the fixed-policy problems obey comparison in the bounded continuous class with the specified boundary conditions.*

The comparison requirement identifies both a function class and a boundary problem. It is not asserted merely because the primitive coefficients are continuous. On a bounded, regular Dirichlet domain, a boundary barrier is one way to verify attainment of the stopping payoff. Reflection requires the corresponding Neumann or oblique derivative condition and a well-posed reflected state process. Unbounded-state applications require growth bounds and localization. These conditions are checked separately from a neural residual.

For a smooth function v , write

$$\mathcal{L}^a v = b(t, x, a)^\top Dv + \frac{1}{2} \text{tr}\{\sigma(t, x, a)\sigma(t, x, a)^\top D^2 v\}, \quad (3)$$

$$\mathcal{H}^a[v] = f(t, x, a, v) + \mathcal{L}^a v, \quad F[v] = -v_t - \sup_{a \in \mathcal{A}(t, x)} \mathcal{H}^a[v]. \quad (4)$$

The value solves $F[V^*] = 0$ with the specified boundary data, in the viscosity sense where classical derivatives do not exist. The diffusion covariance is the positive semidefinite matrix $\sigma\sigma^\top$. No claim that the product of two arbitrary positive semidefinite matrices is symmetric positive semidefinite is needed.

A continuous martingale representation can also be used when its quadratic variation is Markov in an augmented state. If $d\langle M \rangle_s = Q(s, S_s, a_s)ds$, the covariance in (3) becomes $\sigma Q \sigma^\top$. A merely predictable process Q_s is not, without further state augmentation, a deterministic coefficient of a Markov HJB equation. Quadratic variation by itself also does not imply conditionally Gaussian increments. The implementation in this paper uses the Brownian model (1).

2.1. Infinite horizon and economic normalization

For a stationary problem, the boundary-value equation is $\sup_a \{f(x, a, V) + \mathcal{L}^a V\} = 0$. We use an infinite-horizon result only after specifying admissibility, integrability, and the selected utility process. In the time-additive case, a sufficient verification requirement is that the localized Itô identities are uniformly integrable and $\mathbb{E}[e^{-\rho T} |V(S_T)|] \rightarrow 0$ for the policies being compared. Recursive problems require an appropriate utility comparison and transversality condition instead of simply borrowing the additive one.

Cardinal transformations also require care. Multiplying a fixed utility function by a positive constant does not change its maximizer. Adding a function of an *endogenously controlled preference state* generally does. Consumption units, preference normalization, and terminal utility are therefore primitive economic choices in Section F.3.

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The actor $a_\phi(t, x)$ and critic $v_\xi(t, x)$ represent policies and continuation values. An internal actor and an external actor can be separate parameter blocks; their joint action remains the input to the same economic generator. A smooth activation is useful for exact differentiation, while a nonsmooth neural representation can be used with the positive-weight Bellman formulation below.

3.1. Feasibility and boundary data

For a box $[a, \bar{a}]$, a sigmoid map produces interior feasible actions but does not attain its endpoints at finite weights. Binding constraints can instead be represented by a projection onto the closed interval, by an explicit boundary candidate, or by a finite action set including the endpoints. The constrained portfolio calculation uses a projection and can attain a zero risky share exactly. On a simplex, a probability map supplies a feasible interior point and boundary faces must again be considered when certifying the optimum. General constraints require a feasible parameterization, a certified projection, or a constrained action solver. A finite penalty is not an exact feasibility certificate.

On a finite horizon, an elementary hard-terminal architecture is

$$v_{\bar{\zeta}}(t, x) = G(x) + (T - t)N_{\bar{\zeta}}(t, x). \quad (5)$$

This enforces the terminal face, but not automatically a lateral boundary. A compatible lifting G_B and a factor d_B vanishing on all Dirichlet faces give $v_{\bar{\zeta}} = G_B + d_B N_{\bar{\zeta}}$. Alternatively, boundary residuals are trained separately and retained explicitly in the error bound. A reflected preference state requires its own derivative condition; a terminal architecture does not enforce it. For recursive utility, a domain-preserving transformation must additionally keep the continuation value in the domain of the aggregator.

3.2. The block-specific computational graph

Let ν be a specified distribution of state–time points and let sg denote stop-gradient. A critic step minimizes

$$L_C(\bar{\zeta}; \bar{\phi}) = \mathbb{E}_{\nu} \left[-\partial_t v_{\bar{\zeta}} - \mathcal{H}^{\text{sg}(a_{\bar{\phi}})}[v_{\bar{\zeta}}] \right]^2 + L_B(\bar{\zeta}), \quad (6)$$

where L_B is absent only for boundary conditions enforced exactly. The policy, sampling locations, and any rival actions are detached. The value, time derivative, state gradient, and Hessian remain differentiable with respect to the critic parameters.

An actor step minimizes

$$L_A(\phi; \bar{\zeta}) = -\mathbb{E}_{\nu} \mathcal{H}^{a_{\phi}}[\text{sg}(v_{\bar{\zeta}}), \text{sg}(Dv_{\bar{\zeta}}), \text{sg}(D^2v_{\bar{\zeta}})]. \quad (7)$$

All critic derivative channels are fixed in this step. There is no critic gradient from L_A and no actor gradient from L_C . A printed sum $L_A + \omega L_C$ may be a reporting device, but it is not jointly differentiated. The implementation exposes separate functions and regression tests for these derivative paths.

The distinction is substantive: joint differentiation can move the critic away from the policy-evaluation root. Appendix A gives a counterexample, the recursive lifetime policy derivative, and the conditioning requirements. The local Hamiltonian objective is not identified with a lifetime policy gradient under arbitrary state sampling. The evaluated reference policy may remain fixed while a separate candidate

actor is improved at successive work levels. Updating that reference defines a new evaluation target and is recorded explicitly.

3.3. Rollout and costate evaluation

For a frozen policy $a^{(j)}$, a positive-time Euler rollout from (t, y) supplies a sample return $\widehat{V}^{(j)}(t, y)$ and its pathwise derivative $\widehat{p}^{(j)}(t, y)$. Differentiation includes the state dependence of subsequent policy actions. Freezing the policy means freezing its parameters, not deleting that derivative. The targets evaluate the current policy, not the optimal policy, and contain no maximizing action labels. For fixed finite weights, bounded actions and smooth bounded production justify pathwise differentiation of this finite rollout away from the harmless clipping kinks. These are derivatives of the discretized evaluation problem, not asserted unbiased derivatives of its continuous-time limit.

The critic minimizes

$$\widehat{\mathbb{E}} \left[0.05 \{v_\theta - \widehat{V}^{(j)}\}^2 + \|d(D_y v_\theta - \widehat{p}^{(j)})\|_d^2 \right]. \quad (8)$$

The factor d puts normalized marginal values on an order-one scale. The rollout implementation uses two hidden tanh layers and a hard terminal component. Holding its new derivative fixed, the actor then maximizes

$$\widehat{\mathbb{E}} \left[\overline{\log a_\phi} - \frac{\zeta}{2} \bar{a}_\phi^2 - a_\phi \cdot \text{sg}(D_y v_\theta) \right]. \quad (9)$$

Terms independent of actions have been omitted. Consumption is represented directly by $m_0(t) + \epsilon \tanh N_\phi(t, y)$; there is no online Hamiltonian search in its deployment. Value evaluation, costate evaluation, and feasible improvement are distinct operations, with their own stored losses and work counts. The number of critic and actor steps per rollout batch is a declared design parameter. Failure of either numerical block is recorded, not interpreted as successful policy improvement.

The stochastic gradient targets make the economically relevant marginal continuation values visible to the evaluation loss. This is not a new Bellman principle, nor an assumption that derivative regression invariably dominates differential residual fitting. It is a concrete implementation of the NBO evaluation map. The preserved

differential implementation and its two independent trace banks continue to test a different evaluation procedure. Both feed the same economic distinction between policy evaluation and feasible improvement.

3.4. The implemented map and its verification output

Fix the economy, boundary realization, network classes, optimization budget, and verification procedure, denoted collectively by ϑ . Let $\mathcal{P}$ be the admissible feedback policies, Ξ the critic parameter space, and Ω the sampling streams. The implemented NBO is the numerical map

$$\mathcal{N}_\vartheta : \mathcal{P} \times \Xi \times \Omega \longrightarrow \mathcal{V}_\vartheta \times \mathcal{P}_\vartheta \times \mathcal{C}_\vartheta, \quad (a, \xi, \omega) \longmapsto (v, \tilde{a}, C). \quad (10)$$

The first component is approximate policy evaluation; the second is feasible improvement; the third records verification of the resulting pair. A record identifies the model, parameters, domain, boundary faces, evaluation error, maximization error, coverage, discretization errors, and unchecked assumptions. An unavailable bound is not zero. Failure to obtain a useful certificate is a possible output of the map. With its sampling stream fixed, the implementation is deterministic. We do not assert that a network learns an unrestricted Bellman operator on a function space.

The evaluation–improvement structure is an instance of approximate policy iteration, including its partial-evaluation variants (Scherrer et al., 2015). Neural representation does not introduce a new principle of optimality, and monotonicity belongs to the underlying positive-weight scheme, not to gradient descent. The NBO specification joins the block-specific derivative graph to boundary and utility-domain restrictions and to a common economic error account. Sections I.1 and F.3.4 implement the differential and positive-weight versions of this map, respectively, with independent verification of their outputs.

4. VERIFICATION, APPROXIMATION, AND VISCOSITY SELECTION

For a feasible policy a and a smooth candidate v , define

$$r^a[v] = -v_t - \mathcal{H}^a[v], \quad g^a[v] = \sup_{b \in \mathcal{A}} \mathcal{H}^b[v] - \mathcal{H}^a[v] \geq 0. \quad (11)$$

Algorithm 1 Neural Bellman evaluation, feasible improvement, and economic assessment

Require: Economic primitives and information; feasible actions and utility domain; boundary data; evaluation variant; training, validation and stopping rules; independent final assessment.

- 1: Declare the feasible reference policy to be evaluated. Initialize a feasible candidate and a boundary-compatible continuation representation.
 - 2: **for** each permitted evaluation–improvement iteration **do**
 - 3: Freeze the declared reference policy, rivals and evaluation target. Fit only the critic by the declared differential, positive-weight, or rollout rule. The reference may remain fixed across work levels.
 - 4: Freeze the critic and its value/derivative channels. Update only the feasible actor, or apply the declared feasible action solver.
 - 5: Record all evaluation and improvement work, corrections, unsuccessful checks, and the resulting deployment rule.
 - 6: Apply the predetermined validation and stopping rule, using no final-test outcomes.
 - 7: **end for**
 - 8: Freeze the selected policy and continuation object before independent final evaluation.
 - 9: Compute the specified economic endpoints and direct contrasts, with their approximation, implementation and statistical accounts.
 - 10: Return the fitted objects, their implementation and domain, and the complete accuracy and work record.
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The first error concerns evaluation; the second concerns improvement. Their difference is the maximized HJB residual: $F[v] = r^a[v] - g^a[v]$. We keep the two quantities separate because the policy value and its regret depend differently on them.

THEOREM 1—Finite-horizon economic error bound: *Under Assumption 1, suppose a bounded smooth candidate v in the declared utility interval and an admissible Markov policy a satisfy $|r^a[v]| \leq e$, $0 \leq g^a[v] \leq \eta$ throughout the verification domain. Suppose the*

boundary payoff error is at most B . Any reflected boundary condition is imposed exactly.

Set $A_L(s) = (e^{Ls} - 1)/L$ for $L > 0$ and $A_0(s) = s$. Then, for $s = T - t$,

$$|v(t, x) - V^a(t, x)| \leq e^{Ls}B + A_L(s)e, \quad (12)$$

$$|v(t, x) - V^*(t, x)| \leq e^{Ls}B + A_L(s)(e + \eta), \quad (13)$$

$$0 \leq V^*(t, x) - V^a(t, x) \leq 2e^{Ls}B + A_L(s)(2e + \eta). \quad (14)$$

Consequently, exact evaluation, exact boundary data, and a zero feasible-action gap identify the optimal value and an optimal policy.

The theorem replaces an assertion about a global minimizer of a composite loss with a statement about the exact computational objects NBO estimates. It allows errors, does not require that a finite network class contain V^* , and bounds performance against all admissible policies covered by comparison. A small sampled residual is not yet the uniform hypothesis of the theorem. Section A.2.4 addresses that distinction.

4.1. A positive-weight recursive Bellman approximation

Let P_h^a be a nonnegative transition/interpolation operator on a finite state set, with row sums one after adjoining absorbing states. Boundary values are fixed. For a continuation array w , define $\Psi_h^a w$ pointwise as the scalar solution

$$z = P_h^a w + hf(x, a, z), \quad T_h w = \max_{a \in \mathcal{A}_h(x)} \Psi_h^a w. \quad (15)$$

For stationary calculations assume $f_y \leq -\lambda < 0$ on an invariant value domain, and require that all scalar roots remain in that domain. The map $z \mapsto z - hf(x, a, z)$ is strictly increasing with slope at least $1 + \lambda h$. Hence the root is unique and Ψ_h^a is order preserving and a sup-norm contraction with factor $q_h = (1 + \lambda h)^{-1}$. The maximum over feasible actions preserves these properties. A root bracket must contain the root; arbitrary clipping of utility values is not part of the definition.

For a finite horizon, the same implicit construction is used backward from the prescribed terminal array. Under the Lipschitz condition in Assumption 1, $hL < 1$ gives a one-step Lipschitz constant at most $(1 - hL)^{-1}$. Discount contraction is not

assumed for a recursive parameter regime where it has not been verified. For a time-additive problem, the positive-weight backup $hr_0 + e^{-\rho h} P_h^a w$ is also consistent and monotone. It is the backup used by the NDU reference laboratory.

THEOREM 2—Discrete NBO certificate: *Suppose (15) is well defined on an invariant domain, with contraction factor $q_h < 1$. Let v_h^* and v_h^a denote the fixed points of T_h and T_h^a , respectively. For any neural value array v and deterministic feasible policy a , set*

$$\delta = \|v - T_h^a v\|_\infty, \quad \epsilon = \|T_h v - T_h^a v\|_\infty.$$

Then

$$\|v - v_h^a\|_\infty \leq \frac{\delta}{1 - q_h}, \quad \|v - v_h^*\|_\infty \leq \frac{\delta + \epsilon}{1 - q_h}, \quad \|v_h^* - v_h^a\|_\infty \leq \frac{2\delta + \epsilon}{1 - q_h}. \quad (16)$$

The certificate covers the stated finite state and action model. Continuous actions require an additional certified maximization or action-coverage error.

This result applies regardless of how the neural array was obtained. Neural projection need not itself be monotone. What matters is that the output satisfies the monotone Bellman equation to the displayed tolerance. The distinction makes it possible to use flexible representations while retaining a comparison-based acceptance test.

THEOREM 3—Viscosity-consistent neural Bellman approximation: *Consider stationary problems satisfying comparison in a bounded continuous class and the strict utility monotonicity used in Theorem 2. Suppose the positive-weight schemes are stable, locally consistent with the HJB operator, and attain the prescribed boundary data in the limit. The feasible action approximations become dense with vanishing maximization error. If neural arrays and feasible actors satisfy*

$$\delta_h + \epsilon_h = o(h), \quad (17)$$

then their interpolated values converge locally uniformly to the unique viscosity solution. Uniform convergence up to the boundary follows when boundary barriers give uniform boundary attainment. Their discrete policy-value arrays converge to the same limit. No assertion of convergence of the policy functions themselves is required.

Here stability, local consistency, comparison, and boundary attainment are numerical-analysis hypotheses, not consequences of calling a representation neural. The proof is given in the appendix. A finite-horizon version uses backward error accumulation: terminal error tends to zero and the sum of one-step errors tends to zero after multiplication by the one-step Lipschitz factors. This covers finite-horizon recursive problems without imposing stationary discount contraction.

For a diffusion with m Brownian factors, an interior semi-Lagrangian construction uses the $2m$ locations $x + hb \pm \sqrt{mh} \sigma e_j$, each with weight $1/(2m)$. Positive multilinear interpolation preserves monotonicity. For smooth test functions its interpolation contribution is $O(\Delta x^2/h)$; thus $\Delta x^2/h \rightarrow 0$ is a sufficient interior coupling condition. The mean and covariance of the cubature increments match those of the specified Brownian diffusion to first order. Stopping and reflection require consistent boundary transitions. Negative interpolation weights are not permitted in this construction.

Appendix A gives the differential-residual counterexample, independent domain and action coverage, the complete-action envelope, and the guarded finite-model evaluation map. Appendix B proves the results, including finite-horizon error accumulation.

5. CONTINUOUS-ECONOMY GUARANTEES FOR NEURAL CAPITAL POLICIES

The distinction between a nodal certificate and an economic diffusion bound is substantive. This section obtains the latter for the original dense, nonlinear capital economy, without replacing its primitives, restricting the comparison policy class, or extrapolating a sampled residual. The construction uses the curvature of consumption utility to control the gain from an arbitrary adapted deviation. A neural actor is then confined to a known neighbourhood of a feasible schedule. Its approximation class is restricted; the economic optimum against which it is assessed is not.

Write $\bar{z} = d^{-1} \sum_i z_i$, $\|z\|_d^2 = d^{-1} \sum_i z_i^2$, and $\Pi z = z - \bar{z}1$. In the capital model already specified, the state and payoff are

$$dY_i = \{a + \kappa \tanh((BY)_i) - m_i - v^2/2\} dt + \sigma_I dW_i + \sigma_C dW_0, \quad (18)$$

$$J_y(m) = \mathbb{E}_y \left[\int_0^T e^{-\rho t} \{ \overline{\log m_t} + \bar{Y}_t - \frac{\zeta}{2} \bar{m}_t^2 \} dt + e^{-\rho T} \{ \bar{Y}_T - \chi \|\Pi Y_T\|_d^2 \} \right], \quad (19)$$

where $v^2 = \sigma_I^2 + \sigma_C^2$ and each control lies in $[\underline{m}, M]$. Controls are progressively measurable and the state equation is interpreted on $\mathbb{R}^d$. Bounded control and the globally Lipschitz bounded production term yield the required state moments and finite payoffs. The historical parameters are $T = 1$, $\rho = 0.04$, $a = 0.10$, $\kappa = 0.15$, $\sigma_I = 0.15$, $\sigma_C = 0.10$, $\zeta = 0.2$, $\chi = 0.03$, and $[\underline{m}, M] = [0.02, 2]$. The stored coupling matrices are unchanged in dimensions ten and twenty. Dimension fifty uses the same stated matrix-generation rule.

Define

$$w(t) = \frac{1 - e^{-\rho(T-t)}}{\rho} + e^{-\rho(T-t)}, \quad m_0(t) = \frac{2}{w(t) + \sqrt{w(t)^2 + 4\zeta}}. \quad (20)$$

This schedule is strictly interior for the parameters above. It is a feasible benchmark, not an asserted solution of the coupled economy. For the analytical comparison, keep m_0 interior and assume $\rho > 0$, $\zeta, \chi, \kappa \geq 0$, and let $\beta \geq \kappa \|B\|_2$ and $s_0 \geq \|\Pi y\|_d$. Choose $\epsilon \geq 0$ so that $[m_0(t) - \epsilon, m_0(t) + \epsilon] \subset [\underline{m}, M]$ at every date. With $V^*(y) = \sup_m J_y(m)$ over the full adapted admissible class, the schedule gap obeys $0 \leq V^*(y) - J_y(m_0) \leq G_d$. The exact scalar integral defining G_d is (S.14); Appendix C gives its curvature and dispersion constants, Theorem S.4, and the complete proof. This analytical bound supplies a common economic reference. The policy-specific endpoint below measures what the fitted actor adds to that reference.

5.1. The deployed neural map and its numerical guard

The actor is a two-hidden-layer network with the closed-form output map

$$m_i^\phi(t, Y) = m_0(t) + \epsilon \tanh N_{\phi,i}(t, Y). \quad (21)$$

The weights are trained by the same block-specific NBO losses as before. No maximizing policy labels, nodal action table, or online local search enter this actor. Input clipping at very large log-capital values protects numerical evaluation; it does not stop or truncate the economic state. The theorem remains valid for any bounded

proposal followed by the stated output constraint, including this clipped-input network.

Numerical implementation must not replace the exact schedule by an unverified floating-point centre. The deployment routine therefore encloses $m_0(t)$ by $[\ell(t), u(t)]$ using the inherited outward arithmetic kernel and projects the proposal onto

$$[\text{up}(u(t) - \epsilon), \text{down}(\ell(t) + \epsilon)]. \quad (22)$$

This interval lies inside the exact tube. Adjacent clock floats enclose rounding of the input time. An empty interval causes an error; a nonfinite proposal is replaced by its interval midpoint and is not evidence of successful training. This scalar feasibility guard is neither an action-maximization oracle nor a stored policy correction table. Its cost is included in the deployment measurements.

5.2. Policy-Specific Verification after Bellman Improvement

The bound in Theorem S.4 is a useful starting point, but not a measure of what training accomplishes. Every policy in the same tube receives the same allowance, including a poorly trained policy. We now replace that allowance by an independently verified, policy-specific payoff difference. The economic primitives, initial state, and full adapted comparison class in (18)–(19) are unchanged. The implementation to which the new bound applies is specified explicitly below. It is not identified with continuously observed-state feedback.

5.2.1. The deployed controller

Fix a deterministic grid $t_k = kh$, $Nh = T$. The controller has an internal state Z_k , initially y , and observes the Brownian innovations at the grid dates. At date t_k it selects

$$a_k = \mathcal{P}_k\{m_\phi(t_k, Z_k)\}, \quad Z_{k+1} = Z_k + h\{c\mathbf{1} + f(Z_k) - a_k\} + \Sigma\Delta\tilde{W}_k, \quad (23)$$

where $c = a - v^2/2$, $f(z) = \kappa \tanh(Bz)$, and $\mathcal{P}_k$ projects inward onto the declared action interval around $m_0(t_k)$. The vector $\Delta\tilde{W}_k$ clips each standardized innovation at ten before rescaling by $\sqrt{h}$. The numerical implementation and its rounding

convention form part of the controller. Between dates it holds a_k fixed. Its economic state, denoted Y^ϕ , nevertheless satisfies

$$dY_t^\phi = \{c\mathbf{1} + f(Y_t^\phi) - a_k\} dt + \Sigma dW_t, \quad t_k \leq t < t_{k+1}.$$

In particular, the economic Brownian motion and state are not clipped, stopped, or replaced by Z . The controller is adapted to the original innovation filtration and remains feasible. Observing innovations is an implementation assumption, not a conclusion inferred from a state-only observation model. The continuously evaluated absolute-bound study policies remain separate objects in the supplement.

This specification has an important analytical consequence. When the numerical state and the economic state are compared, they use exactly the same chosen actions. Their state discrepancy therefore depends on the smoothness of the economic drift, not on a global Lipschitz bound for the learned network. The following construction uses this fact rather than inserting sampled residuals into a uniform verification theorem.

5.2.2. An exact improvement identity

Write $W(t) = e^{-\rho t}w(t)$ and define the nonnegative consumption deficit

$$R_t(m) = \log m_0(t) - \overline{\log m} + w(t)\{\bar{m} - m_0(t)\} + \frac{\zeta}{2}\{\bar{m}^2 - m_0(t)^2\}. \quad (24)$$

The first-order identity $m_0^{-1} = w + \zeta m_0$ and concavity of the logarithm imply $R_t(m) \geq 0$. The exact mean-state identity in the proof of Theorem S.4 gives

$$\begin{aligned} \Delta_\phi := J_y(a^\phi) - J_y(m_0) &= \mathbb{E} \int_0^T W(t) \overline{f(Y_t^\phi) - f(Y_t^0)} dt - \mathbb{E} \int_0^T e^{-\rho t} R_t(a_t^\phi) dt \\ &\quad - \chi e^{-\rho T} \mathbb{E}\{\|\Pi Y_T^\phi\|_d^2 - \|\Pi Y_T^0\|_d^2\}. \end{aligned} \quad (25)$$

Thus state-dependent consumption has to earn its utility-curvature cost through production or terminal dispersion. The identity is also an effective variance reduction: the common mean-state martingale cancels before simulation statistics are formed. Positive neural output is not itself evidence of positive economic improvement.

For each cell, calculate the scalar integrals

$$\begin{aligned} A_k &= \int_{t_k}^{t_{k+1}} e^{-\rho t} dt, & B_k &= \int_{t_k}^{t_{k+1}} W(t) dt, \\ C_k &= \int_{t_k}^{t_{k+1}} e^{-\rho t} \{\log m_0(t) - w(t)m_0(t) - \zeta m_0(t)^2/2\} dt. \end{aligned}$$

The consumption-deficit integral for a held action is exactly $C_k - A_k \overline{\log a_k} + B_k \bar{a}_k + \zeta A_k \bar{a}_k^2/2$. Consequently, consumption need not be approximated by a left-point utility sum. Let $X_{\phi,h}$ be the paired statistic obtained from (25) by using the internal Euler states in the production and terminal terms and these integrated consumption terms. The reference internal state uses the integrated common action $\int_{t_k}^{t_{k+1}} m_0(t) dt$. All scalar integral enclosures and their midpoint errors are saved.

5.2.3. Transferring the computed payoff to the economy

The statistic $X_{\phi,h}$ is evaluated for the held controller in (23), using the same actions when comparing its internal state with its economic diffusion. Proposition S.5 in Appendix C bounds $|\Delta_\phi - \mathbb{E}X_{\phi,h}|$ by the saved allowance $E_{\phi,h}^{\text{all}}$. The exact-arithmetic diffusion allowance is $E_\epsilon(h) + E_0(h)$, with E_ϵ defined in (S.23). The appendix gives the additional clipped-innovation, forward-arithmetic, and enclosed-integration terms. The $O(h)$ diffusion bound uses bounded actions, the global drift and derivative bounds of the specified hyperbolic-tangent production function, and additive diffusion; it requires no derivative bound for the trained network.

5.2.4. A post-training certificate

Use the deterministic clipping threshold b_ϕ and expectation-tail allowance τ_ϕ constructed in Appendix C.4. They account for the unbounded Gaussian terminal-dispersion term and are not estimated from the final sample range. Let $X_{\phi,h}^c$ be $X_{\phi,h}$ clipped to $[-b_\phi, b_\phi]$, so that $|\mathbb{E}X_{\phi,h} - \mathbb{E}X_{\phi,h}^c| \leq \tau_\phi$.

For n fresh independent paths, let $\bar{X}_\phi^c$ and s_ϕ^2 be the clipped empirical mean and unbiased variance. Fix a family of at most F one-sided statements before inspecting this noise bank, and set $\ell = \log(2F/\alpha)$. Define

$$L_\phi = \bar{X}_\phi^c - \sqrt{2s_\phi^2 \ell / n} - \frac{14b_\phi \ell}{3(n-1)} - E_{\phi,h}^{\text{all}} - \tau_\phi. \quad (26)$$

In the implementation the empirical mean is enclosed downward and the variance and every subtracted quantity upward. The upper payoff endpoint uses the same terms with reversed signs. The factor F counts both tails when both are reported.

THEOREM 4—Policy-specific continuous-time certificate: *Condition on the training and validation data, and suppose the specified final simulation paths are independent draws from the declared innovation law. Under the arithmetic contract and transfer assumptions above, with probability at least $1 - \alpha$, simultaneously for all declared policies and initial states,*

$$J_y(a^\phi) - J_y(m_0) \geq L_\phi, \quad 0 \leq V^*(y) - J_y(a^\phi) \leq G_d - L_\phi. \quad (27)$$

The optimum V^ still ranges over all adapted feasible controls in the original continuous economy, not merely the networks, schedules, or sampled controllers used in the experiment. Common shocks may be reused across policies; independence across those policies is not required. The guarantee is pointwise in each declared initial state and conditional on training, rather than uniform on an uncountable state domain.*

PROOF: Apply the empirical Bernstein inequality of [Maurer and Pontil \(2009, Theorem 4\)](#) to $(X_{\phi,h}^c + b_\phi) / (2b_\phi)$ and use a union bound over the declared one-sided statements. Rescaling gives the first two error terms in (26). The clipping expectation bound and Proposition [S.5](#) then give the first inequality in (27). Subtract it from the deterministic anchor bound ([S.14](#)). Feasibility yields nonnegative regret. An observed negative upper regret endpoint would indicate failure of the confidence event or of an implementation assumption, not a negative optimality gap. *Q.E.D.*

Unlike $G_d + D_{d,\epsilon}$, the new endpoint depends on the deployed policy through its paired outcomes and sample variance. It can distinguish seeds and unsuccessful policies in the same architectural tube. When $L_\phi > 0$, the post-training guarantee is tighter than the analytical schedule’s guarantee. This is a statement about the verified policy, not a deterministic promise that every training run improves welfare. The concentration inequality, the identity that transfers a lower improvement bound into a regret bound, and conservative policy improvement are established ideas ([Kakade and Langford, 2002](#), [Pirotta et al., 2013](#), [Scherrer et al., 2015](#)). The

contribution here is their operational connection to the continuous capital economy through (25) and the transfer bound (S.23), without a state-covering certificate or a global bound on the trained network's derivatives.

6. BELLMAN EVALUATION AND DIRECT METHOD COMPARISONS

The NBO map has two separate empirical obligations. Its evaluation block must produce a useful continuation-value derivative, and the resulting policy must justify the resources used to construct it. Verification of a policy does not by itself answer the second question. We therefore connect evaluation error to Hamiltonian error and compare candidate-generation methods by a direct paired payoff statistic. The economic problem and the feasible competitor class remain unchanged.

6.1. What the critic contributes

Use the normalized inner product $\langle x, z \rangle_d = d^{-1} x^\top z$ and norm $\|x\|_d^2 = \langle x, x \rangle_d$. At a state of the capital economy, set $q = dD_y V^\pi$. The part of the Hamiltonian depending on consumption is

$$H_q(m) = d^{-1} \sum_i \log m_i - \frac{\zeta}{2} \bar{m}^2 - \langle q, m \rangle_d.$$

On the original action box, and on any convex tube contained in it, H_q is μ -strongly concave for $\mu = \bar{m}_{\text{box}}^{-2}$. Here $\bar{m}_{\text{box}}$ denotes the primitive upper consumption-share bound, not the cross-sectional mean. Let m_q maximize H_q , let $\hat{q}$ be a critic costate, and let $\tilde{m}$ be within δ of the maximum of $H_{\hat{q}}$ over the same action set.

PROPOSITION 5—Evaluation error and feasible improvement: *Under the preceding conditions,*

$$H_q(m_q) - H_q(\tilde{m}) \leq \left(\sqrt{\delta} + \frac{\|q - \hat{q}\|_d}{\sqrt{2\mu}} \right)^2. \quad (28)$$

For a random rollout costate Z and a square-integrable predictor $g(X)$,

$$\mathbb{E} \|Z - g(X)\|_d^2 = \mathbb{E} \|Z - \mathbb{E}[Z | X]\|_d^2 + \mathbb{E} \|\mathbb{E}[Z | X] - g(X)\|_d^2. \quad (29)$$

These statements do not identify the costate of an Euler rollout with the continuous-time costate; a discretization or evaluation bias must be added when that identification is required.

Equation (28) identifies a reason to regress costates rather than only value levels: Hamiltonian loss is quadratic in a suitably measured costate error when the actor solves its improvement problem. Equation (29) identifies both the variance reduction available from conditional prediction and its regression bias. Neither identity promises that a trained critic reduces mean-square error relative to raw rollout derivatives. The existing comparison therefore includes raw-costate improvement without a learned critic, value-only and costate-only critics, changes in block-update counts, and direct policy optimization. These are substantive alternatives, not alternative labels for NBO. The strong-concavity and conditional-projection arguments are established mathematical principles; their role here is to specify a testable evaluation-improvement mechanism, not to claim a new principle of policy iteration.

The continuous-time performance-difference account in Proposition S.3 integrates evaluation and action errors under the candidate’s occupation law. Its full statement, proof, and alternative residual-to-costate conditions are in the supplement. Training-design prediction errors alone do not meet that account. The current finite-step mechanism below instead states its continuation target exactly and carries its welfare comparison to the continuous economy through the separate payoff-transfer bound.

6.2. Direct paired economic comparisons

Let a and b be two fitted sampled controllers, fixed before the final simulation bank. For each independent final path, compute their schedule-relative statistics $X_{a,h}$ and $X_{b,h}$ with exactly the same initial state and Brownian increments, and form

$$D_{ab,h} = X_{a,h} - X_{b,h}.$$

The common schedule’s numerical trajectory cancels. The implementation checks equality of its terminal arrays and the initial-state and innovation hashes. The

deterministic transfer allowance is

$$E_{ab,h} = E_{a,h}^{\text{policy}} + E_{b,h}^{\text{policy}} + E_{a,h}^{\text{stat}} + E_{b,h}^{\text{stat}},$$

with outward arithmetic and a separate subtraction-roundoff cushion. In particular, it does not add the schedule transfer error twice. If b_a, b_b are the two clipping thresholds and τ_a, τ_b their expectation-tail allowances, use $b_{ab} = b_a + b_b$ and $\tau_{ab} = \tau_a + \tau_b$.

THEOREM 6—Simultaneous method and population comparisons: *Condition on all fitting and validation data. Suppose the final paths follow the declared independent sampling law and the conditional arithmetic and diffusion-transfer contract of Theorem 4. Apply the empirical Bernstein construction in (26) directly to $D_{ab,h}$, with threshold b_{ab} and allowances $E_{ab,h}, \tau_{ab}$. The resulting endpoints L_{ab}, U_{ab} satisfy*

$$L_{ab} \leq J(a) - J(b) \leq U_{ab}$$

simultaneously with the other declared one-sided statements at their joint confidence level. A difference of two schedule-relative lower endpoints is not used.

The same construction applies to $J_{\mathcal{P}}(a) = \sum_j p_j J_{y_j}(a)$ for a prespecified finite initial-state population $\mathcal{P}$. Draw its profile independently on every final path, average the conditional deterministic transfer bounds with the exact population weights, and use a common clipping threshold and tail bound valid for all profiles. The resulting guarantee is integrated over $\mathcal{P}$; it is not uniform over a continuum of initial states or over unobserved training seeds.

The comparison does not require NBO to dominate a direct method. It makes that empirical question answerable by the quantity actually of interest. A lower endpoint below zero is inconclusive about superiority unless the upper endpoint is also below zero. Certified improvement over the schedule and certified improvement over another trained policy remain different statements. Likewise, a smaller policy-regret upper endpoint need not equal the unknown actual regret.

6.3. An operational evaluation–improvement account

The costate in a rollout is the derivative of a specified continuation problem. Its approximation error has to be measured where the new policy uses that continuation. We now make both objects explicit. The construction uses the capital economy, including its dense production interaction, common and idiosyncratic shocks, consumption curvature, and terminal dispersion preference. It also accommodates a critic fitted on a coarser grid than the one used for assessment.

Fix the assessment grid $t_k = kh$, $Nh = T$. Write $\pi_k = \bar{M}_k/h$, where $\bar{M}_k$ is the disclosed numerical representative of $\int_{t_k}^{t_{k+1}} m_0(t)dt$. In the following finite problem, these representatives are fixed real numbers. They specify a feasible, state-independent reference. The internal transition and transformed reward are

$$Z_{k+1} = Z_k + h\{c\mathbf{1} + f(Z_k) - a_k\} + \Sigma\sqrt{h} \text{clip}_z(G_k), \quad G_k \sim N(0, I_{d+1}), \quad (30)$$

$$\ell_k(z, a) = B_k \overline{f(z)} + A_k \overline{\log a} - B_k \bar{a} - \frac{\zeta A_k}{2} \bar{a}^2, \quad g_N(z) = -\chi e^{-\rho T} \|\Pi z\|_d^2. \quad (31)$$

Here A_k, B_k are the integrated weights defined above; all discounting is included in them. Let $\mathcal{R}_h(a)$ be the expected sum of these rewards and g_N . The candidate may depend on its entire observed innovation history. Its actions and the reference belong to the same declared convex action box $\mathcal{A}_k$ at each date.

Let V_k^π be the value of this finite problem under π , and define its continuation after choosing the deterministic drift by

$$S_k^\pi(x) = \mathbb{E}[V_{k+1}^\pi\{x + \Sigma\sqrt{h} \text{clip}_z(G_k)\}], \quad q_k^\pi = dD_x S_k^\pi. \quad (32)$$

The argument x is the conditional mean before the current innovation. The current reward ℓ_k is excluded from S_k^π ; subsequent rewards start at $k+1$ on the same global grid. In particular, $q_{N-1}^\pi(x) = -2\chi e^{-\rho T} \Pi x$. Smoothness of f , bounded actions, finite moments, and the quadratic terminal reward justify the derivatives and expectations in this finite problem.

After fitting, freeze an arbitrary continuously differentiable scalar predictor $\hat{S}_k$ and its costate $\hat{q}_k = dD\hat{S}_k$. Set

$$x_b(z) = z + h\{c\mathbf{1} + f(z) - b\}, \quad \hat{Q}_k(z, b) = \ell_k(z, b) + \hat{S}_k\{x_b(z)\}.$$

For U uniform on $[0, 1]$, independent of the candidate's past, the *action bridge* is $X_{k,U} = x_{\pi_k}(Z_k) + U\{x_{a_k}(Z_k) - x_{\pi_k}(Z_k)\}$. Define the following quantities under the candidate's internal-state law:

$$\begin{aligned}\mathcal{M}_h &= \sum_{k=0}^{N-1} \mathbb{E} [\hat{Q}_k(Z_k, a_k) - \hat{Q}_k(Z_k, \pi_k)], \\ \mathcal{A}_h &= \sum_{k=0}^{N-1} h \mathbb{E} \|a_k - \pi_k \mathbf{1}\|_d^2, \quad \mathcal{E}_h = \sum_{k=0}^{N-1} h \mathbb{E} \|\hat{q}_k(X_{k,U}) - q_k^\pi(X_{k,U})\|_d^2.\end{aligned}\tag{33}$$

$\mathcal{M}_h$ records the improvement predicted by the fitted scalar continuation, $\mathcal{A}_h$ records the size of the implemented change, and $\mathcal{E}_h$ measures the error that can affect this particular change. Training-state accuracy alone does not identify $\mathcal{E}_h$.

PROPOSITION 7—Finite Bellman bridge: *If the displayed quantities are integrable, then*

$$\mathcal{R}_h(a) - \mathcal{R}_h(\pi) \geq \mathcal{M}_h - \sqrt{\mathcal{A}_h \mathcal{E}_h}.\tag{34}$$

The assertion holds for every feasible candidate. It requires neither exact maximization of $\hat{Q}$ nor a continuous-time costate approximation. A predictor fitted on another mesh enters $\mathcal{E}_h$ with its full discrepancy from the assessment-grid continuation.

The proposition follows from the finite-horizon performance-difference identity and a line integral of the costate error along each action bridge. It is useful even when action optimization is incomplete. The next result additionally quantifies that incompleteness with a finite, checkable calculation.

PROPOSITION 8—A feasible-actor account: *Suppose $\hat{S}_k$ is twice continuously differentiable and $dD^2\hat{S}_k(x) \preceq \Lambda_k I_d$ everywhere. Let $m_{+,k}$ be the upper endpoint of $\mathcal{A}_k$, and put*

$$\Gamma_k = h^2 \Lambda_k - A_k / m_{+,k}^2, \quad g_{k,i} = A_k \{a_{k,i}^{-1} - \zeta \bar{a}_k\} - B_k - h \hat{q}_{k,i} \{x_{a_k}(Z_k)\}.$$

An upper bound on the actor's $\widehat{Q}_k$ maximization gap is

$$\delta_k = \max_{b \in \mathcal{A}_k} \left\{ \langle g_k, b - a_k \rangle_d + \frac{\Gamma_k}{2} \|b - a_k\|_d^2 \right\}. \quad (35)$$

This is a separable scalar calculation. If $\Gamma_k < 0$, its maximizer is the coordinate projection of $a_k - g_k/\Gamma_k$. If $\Gamma_k \geq 0$, one compares the two endpoints in each coordinate. Thus (35) remains valid when global strong concavity of the learned continuation problem cannot be verified.

If $\mu_0 = \min_k (-\Gamma_k/h) > 0$ is verified and $\mathcal{D}_h = \sum_k \mathbb{E} \delta_k$, then

$$\mathcal{R}_h(a) - \mathcal{R}_h(\pi) \geq \frac{1}{2} \mathcal{M}_h - \mathcal{D}_h - 2\mathcal{E}_h/\mu_0. \quad (36)$$

For exact $\widehat{Q}$ maximizers, $\mathcal{D}_h = 0$ and the coefficient of $\mathcal{E}_h/\mu_0$ improves from two to one. These conclusions apply to the actual feasible action; a small training loss is not substituted for δ_k .

Independent occupation and continuation samples. Draw an independent candidate trajectory, K uniformly from $\{0, \dots, N-1\}$, and U uniformly from $[0, 1]$. At its bridge state, draw two independent continuation banks Z_1, Z_2 with conditional mean $q_K^\pi(X_{K,U})$. A bank may contain antithetic pairs; independence is required between the two banks, not between the members of an antithetic pair. The statistics

$$\begin{aligned} M &= N \{ \widehat{Q}_K(Z_K, a_K) - \widehat{Q}_K(Z_K, \pi_K) \}, & A &= T \|a_K - \pi_K \mathbf{1}\|_d^2, \\ E &= T \langle \widehat{q}_K(X_{K,U}) - Z_1, \widehat{q}_K(X_{K,U}) - Z_2 \rangle_d, & D &= N \delta_K \end{aligned} \quad (37)$$

have expectations $\mathcal{M}_h, \mathcal{A}_h, \mathcal{E}_h, \mathcal{D}_h$. Individual values of E may be negative. Replacing them by their positive parts would change the estimand. Moreover, $E^{\text{raw}} = T \|Z_1 - Z_2\|_d^2/2$ estimates the integrated mean-square error of one raw bank on this same occupation law. A confidence bound for $E - E^{\text{raw}}$ therefore assesses the critic's reusable costate prediction against a direct bank with the declared number of rollouts. It concerns this evaluation subproblem; the direct method comparison continues to use the returned policies' independently assessed payoffs.

The supplement gives explicit clipping ranges and expectation allowances for these statistics. They exploit the common quadratic terminal component and anti-

thetic cancellation. They do not use an empirical maximum as a population bound. Independent, equally sized samples for every member of a declared finite stream distribution may be pooled using the maximum proved range and the mean numerical and tail allowances. The resulting target is the uniform mean over all declared streams.

THEOREM 9—From fitted continuation to continuous economic gain: *Condition on the fitted objects and their selection. Suppose a fresh, simultaneous assessment gives $L_M \leq \mathcal{M}_h$, $\mathcal{A}_h \leq U_A$, $\mathcal{E}_h \leq U_E$, and, when used, $\mathcal{D}_h \leq U_D$. Include the disclosed arithmetic and state-representation errors in these endpoints. Define the deterministic holding deficit*

$$D_{\text{hold}} = \sum_k \left\{ C_k - A_k \log \pi_k + B_k \pi_k + \zeta A_k \pi_k^2 / 2 \right\} \geq 0.$$

For the sampled continuous-economy implementation of Proposition 5.5, let $E_{a,h}^{\text{all}}$ be its saved paired payoff-transfer allowance. Then, on that same event,

$$J(a) - J(m_0) \geq L_M - \sqrt{\max(0, U_A) \max(0, U_E)} - D_{\text{hold}} - E_{a,h}^{\text{all}}. \quad (38)$$

When the positive curvature modulus in Proposition 8 is verified, one may also use $L_M/2 - U_D - 2 \max(0, U_E) / \mu_0$ before subtracting the same two allowances. Taking the larger of these two bounds spends no additional probability. Subtracting a valid lower gain bound from the analytical reference regret bound gives a bound against all adapted feasible controls in the original economy.

This account separates three observable economic requirements: a continuation that predicts a useful feasible change, an error small enough under the candidate's occupation law, and a sufficiently accurate implementation. The finite Bellman identity is exact; the transfer of payoffs to the diffusion is quantitative. Neither equality of discrete and continuous costates nor global concavity of the full adapted-control problem is assumed.

6.4. Information and the implemented decision rule

The economic institution is a controller that receives capital observations at specified dates, stores an internal state, and holds the selected consumption share until the next decision date. Its economic primitive is the original capital diffusion. Its internal numerical state need not equal the physical state. When the controller uses private randomization, that randomization is part of its specified information and is included in the law of the evaluated policy.

The exact-history representation in Proposition S.11 assumes continuous noiseless observation of every capital coordinate, known coefficients and actions, exact drift integration, and an independent private Brownian motion. Under those assumptions the observation history supplies a Brownian representation with the law required by the innovation-driven controller. It does not recover the original latent null-space shock path by path. Proposition S.12 states why the added independent randomization leaves the economic optimum unchanged. These are representation results for a randomized history-dependent policy.

Finite observations require a different implementation account. Proposition S.13 bounds the difference between the measured-state controller and its specified protected parent, retaining sensor error, the internal recurrence, action-map error, clipping, and arithmetic. Its allowance belongs to that parent and that time grid. The empirical comparison reports it alongside the gain measured for the same parent and population; a certificate for a different mesh is not substituted. Complete assumptions and proofs are in Appendix H.

6.5. When is state-dependent management worth its fee?

To give the policy differences a decision interpretation, consider a proportional resource cost $\tau \in [0, 1)$ for administering a state-dependent consumption rule. Gross withdrawals from capital remain $m_i e^{Y_i}$, and the service absorbs $\tau m_i e^{Y_i}$. Household consumption is $(1 - \tau) m_i e^{Y_i}$. Production, gross depletion, the gross-withdrawal adjustment cost, and terminal capital are unchanged. This is a self-financed resource experiment rather than an externally financed utility supplement. It changes the specified fee parameter, not the model used to train or benchmark the gross policy.

Let $A_T = \int_0^T e^{-\rho t} dt$. Log felicity gives the exact identity

$$J_{\mathcal{P}}^{\tau}(a) - J_{\mathcal{P}}(m_0) = J_{\mathcal{P}}(a) - J_{\mathcal{P}}(m_0) + A_T \log(1 - \tau). \quad (39)$$

Hence a positive certified improvement L_a supports every fee τ satisfying

$$\tau < 1 - \exp(-L_a / A_T).$$

A negative upper endpoint after the fee rules out an improvement for that candidate under the confidence event; an interval spanning zero is inconclusive. The fee grid is fixed before the final simulation. We do not equate wall-clock cost to a monetary fee or call these deliberately stylized balance sheets a calibrated household population. The experiment instead states precisely which adoption decisions are supported by the verified utility differences.

7. ECONOMIC APPLICATIONS OF THE BELLMAN STRUCTURE

The motivating economic problem combines external choices with the costly adjustment of preferences. Recursive utility changes how a given policy is evaluated; preference adjustment adds an internal choice to its Hamiltonian; and strategic interaction changes whose deviations define improvement. These distinctions determine the economic interpretation of NBO. The capital economy studied below provides a setting in which policy-specific verification can be carried through to a high-dimensional diffusion. The preference, temporal-self, and game applications establish the corresponding economic objects and their own verification results.

7.1. Costly preference adjustment and portfolio choice

Let u denote risk aversion and X financial wealth. The household chooses consumption c , portfolio share p , and preference-adjustment effort θ . The preference state is reflected on a declared interval $[\underline{u}, \bar{u}] \subset (1, \infty)$, and wealth follows the portfolio diffusion:

$$\begin{aligned} du &= \theta dt + \sigma_u dW^u + dL^{\underline{u}} - dL^{\bar{u}}, \\ dX &= \{[r + p(\mu - r)]X - c\} dt + p\sigma_X X dW^X, \quad d\langle W^u, W^X \rangle_t = \varrho dt. \end{aligned} \quad (40)$$

For a fixed consumption unit $\bar{c} > 0$, flow utility is

$$U(c, u) = \frac{1}{2}k\theta^2, \quad U(c, u) = \frac{(c/\bar{c})^{1-u}}{1-u}.$$

The objective discounts both felicity and adjustment costs at rate ρ and includes the specified wealth-exit and terminal payoff. The baseline uses $\bar{c} = 1$. Because the household controls u , the cardinal normalization across preference states is a primitive: adding a function of u changes the incentives to adjust preferences. The reflected state, positive consumption domain, and stopping data give the evaluation equation its economic and mathematical domain.

At an interior optimum with $V_X > 0$ and $V_{XX} < 0$, the three policy conditions are

$$\begin{aligned} c^* &= \bar{c}(\bar{c}V_X)^{-1/u}, & \theta^* &= V_u/k, \\ p^* &= -\frac{V_X(\mu - r)}{XV_{XX}\sigma_X^2} - \frac{\varrho\sigma_u V_{uX}}{XV_{XX}\sigma_X}. \end{aligned} \quad (41)$$

Box constraints require their corresponding feasible maximizers. The cross derivative in the portfolio rule links the external risk choice to the internal preference state. Its sign is an equilibrium object to be determined by evaluation. The adjustment rule likewise relates effort to the shadow value of changing preferences. A higher adjustment cost lowers the value of every fixed policy and hence the optimized value; whenever an envelope derivative exists,

$$\partial_k V^k(t, u, X) = -\frac{1}{2}\mathbb{E}_{t,u,X}^{a^k} \int_t^\tau e^{-\rho(s-t)} \theta_s^2 ds \leq 0. \quad (42)$$

This comparative static holds with the feasible class and stopping payoff fixed.

The numerical application implements these choices jointly. The finite-action study evaluates complete policies by independent backward recursion. The continuous-action study trains the actor through its own payoff and verifies its hybrid policy with a complete action-box enclosure. Fresh training on nested grids separates the effect of resolution from the evaluation of a frozen coarse policy. A model-specific compensating-consumption calculation translates its cardinal utility bounds into an explicit transfer experiment. Supplementary Sections [F.3](#),

[F.3.4](#), and [F.8](#) give the full model, all recorded policies, refinement evidence, and compensation account.

7.2. Recursive utility and temporal selves

Recursive utility enters policy evaluation through the policy's own continuation utility. With $a = 1 - 1/\psi$, $b = 1 - \gamma$, and $bV > 0$, the Epstein–Zin aggregator used here is

$$f(c, V) = \frac{\rho}{a} \{c^a (bV)^{1-a/b} - bV\}. \quad (43)$$

The invariant utility interval and boundary payoff are part of the evaluation problem. Supplementary Section [F.2](#) derives the homothetic policy conditions and verifies the matching-bequest experiment, including the barriers that keep every policy utility in the aggregator's domain. This application tests recursive evaluation while maintaining the same separation between value approximation and feasible action improvement.

Time inconsistency requires a different improvement criterion. For the continuous discount kernel

$$h(s) = \beta e^{-\rho s} + (1 - \beta) e^{-(\rho + \kappa)s}, \quad 0 < \beta \leq 1, \quad \kappa > 0,$$

two critics evaluate the same future policy at the two discount rates. Writing these values as W_1, W_2 , the current self uses $V = \beta W_1 + (1 - \beta) W_2$ and chooses an instantaneous deviation by maximizing $u(a) + \mathcal{L}^a V$. Supplementary Section [F.4](#) states the complete component equations and derives the spike-deviation condition. In the homothetic consumption economy, the equilibrium consumption share is the unique root in its admissible interval, and stronger present bias raises that share. Independently optimized finite-duration deviations check the limiting equilibrium condition. Thus the actor's objective follows from the temporal equilibrium concept, while the critics retain their policy-evaluation role.

7.3. Capital investment and strategic deviations

In the dynamic Cournot application, firm i chooses investment in its own capital, with

$$dK^i = (I^i - \delta K^i) dt + \sigma_K K^i dW^i, \quad q_i = AK^i.$$

Its flow profit is $P_N(K)q_i - I^i - b_i(I^i)^2/2$, where the market-size normalization in P_N is specified for each economy. Given rivals' policies, a critic evaluates the resulting own-firm payoff and the actor changes only its own action. Consequently, the relevant economic error is the gain from a complete unilateral deviation against the frozen rival profile.

Supplementary Sections F.5 and F.5.3 specify the boundaries, investment sets, and market normalizations, and report independent full dynamic best responses to every principal finite neural profile. The resulting profit-unit bounds include all dates and states of the stated finite game. They also cover randomized and history-dependent unilateral deviations against those fixed Markov rivals, by finite dynamic programming. The record reports the cost of the exhaustive training oracle and any nonzero one-step equilibrium defects along with the final exploitability calculation.

These applications give the common method a precise economic interpretation. A recursive critic evaluates a utility process; preference and capital actors allocate resources and adjustment effort; a temporal actor satisfies a spike condition; and a strategic actor is assessed against unilateral best responses. Each application supplies the boundary conditions, approximation class, and error account required by its own economic question.

8. COMPUTATIONAL DESIGN AND THE EXISTING COMPARISON

The numerical question is whether the learned evaluation block improves the economic accuracy or complete work of the NBO map. All comparisons identify the economic model, feasible actions, evaluation target, selected policy, final payoff law, and work boundaries. A policy-specific confidence interval conditions on its fitted objects. A statement about a training method additionally specifies the distribution

of training randomness and its sampling uncertainty. An interval containing zero establishes neither superiority nor practical equivalence.

8.1. *The executed capital comparison*

The source-bound reference study contains ninety primary fitted policies in dimensions ten, twenty, and fifty. Its primary fitting design uses an announced twenty-second budget, completes the current iteration, and selects the checkpoint using validation data. Each final primary row uses 8,192 independent paths and 1,024 time cells. The comparison concerns a fixed nine-profile population and a separate origin calculation. The simultaneous inference construction covers the declared finite family of selected policies and profiles.

Every fitted policy in the reported primary schedule comparisons has a positive simultaneous lower gain endpoint. For the prespecified population, all thirty NBO policies have this property. These are improvements over the analytical schedule; the direct method comparisons have a different conclusion. Table I retains the findings that motivate the current evaluation experiment.

The negative sample means do not prove that Raw dominates NBO, because the simultaneous direct intervals remain inconclusive. The mechanism comparisons nevertheless provide no positive evidence for the fitted critic in that design. They cannot be converted into a favorable mechanism result by observing that the common verifier accepts NBO policies. Likewise, the fixed-work experiment matches simulator visits and checkpoints but has different optimizer update counts; it does not establish equal total computation.

8.2. *Accuracy objects and economic scale*

The original finite-population NBO regret upper bounds are approximately 0.02097, 0.02847 and 0.03039 in dimensions ten, twenty and fifty. The corresponding mean schedule improvements are approximately 0.00147, 0.00135 and 0.00161. The regret bounds combine the analytical anchor gap with the measured improvement. They are reported as absolute upper bounds, without interpreting them as tight high-dimensional HJB errors.

TABLE I
EXECUTED COMPARISON OF THE LEARNED BELLMAN BLOCK

Object	Recorded result
Primary NBO-minus-DPO/affine intervals	All 120 contain zero
Fixed-work direct method intervals	All 42 contain zero
Principal NBO-minus-Raw payoff means	All 12 are negative; intervals contain zero
Raw versus NBO, dimension-checkpoint cells	Higher Raw mean and lower Raw fitting clock in all 6 cells
Critic versus Raw costate MSE	Higher critic MSE in all 6 principal panels; ratio about 1.06–1.99
Critic-greedy versus Raw Hamiltonian gap	Higher critic-greedy gap in all 6 panels

Note: The primary and fixed-work intervals are policy-payoff comparisons conditional on the fitted objects. The twelve means refer to the principal Ubuntu 24 dimension, stream and checkpoint combinations. The six mechanism panels use held-out training-design states and a common independent finite-rollout reference; they do not instantiate a continuous-time candidate-occupation welfare bound. The complete original rows, arrays, work measurements and source identities remain in the reproduction record.

The independent scalar computation uses a monotone implicit HJB scheme and full-box action maximization on four original grids. At initial log capital zero, the finest domain-six NBO loss is about 3.44×10^{-6} , compared with 1.16×10^{-4} for direct policy optimization. At initial state one, the corresponding losses are about 1.05×10^{-4} and 5.02×10^{-5} . The ordering is therefore state dependent. These are finite-grid losses of Markov feedbacks in the scalar model; the widening experiment changes both domain and spatial spacing. The residual, refinement and five-state accounts remain in Appendix I.

The balance-sheet profiles and the resource fee in (39) are transparent model experiments. They are not an empirical calibration. The fee transformation reports what administration cost a verified utility gain can support under the specified log-felicity resource account. It does not provide an additional independent source of evidence for the critic.

8.3. *Work and comparison semantics*

The original time-budget and fixed-work records retain fitting clocks, simulator visits, optimizer updates, peak process memory, selected checkpoints and verification components. Their verification clock begins after weight loading and deterministic-constant construction. Their prefix frontier is retrospective. Those quantities remain useful descriptions of the original computation, with those boundaries.

The current method-level comparison specifies its training-randomness estimand, economic target, actual stopping rule, complete cost boundary, and final inferential object before its confirmatory evaluation. It must charge failed checks and retain unattained targets. A direct HJB or policy-iteration comparator isolates the role of a separately learned actor and continuation object under a shared model and verifier. A classical low-dimensional method serves as a policy generator as well as an independent reference. The following study gives the concrete protocol and its executed results.

9. EVALUATION AND ECONOMIC PERFORMANCE

The continuation-learning comparison and the finite-observation exercise address separate questions in the capital economy. The former assesses continuation evaluation and feasible improvement under its stated volatility parameters and complete method distribution. The latter evaluates the original saved policies with a protected parent at their original volatility parameters. Their controller identities, information sets, and probability accounts remain distinct.

9.1. *The capital experiment*

The experiment uses the capital state equation and objective in (18)–(19), in dimensions ten and fifty. It changes the volatility parameters to $\sigma_I = 0.6$ and $\sigma_C = 0.3$; the remaining primitives are $T = 1$, $\rho = 0.04$, $a = 0.10$, $\kappa = 0.15$, $\zeta = 0.2$, $\chi = 0.03$, and $[m, M] = [0.02, 2]$. The initial population assigns equal mass to nine profiles with means $-0.5, 0, 0.5$ and cross-sectional spreads $0, 0.25, 0.5$. The methods share the declared radius-0.1 feasible tube, coupling matrix, initial population, and payoff

evaluator. This is a parameter regime of the same economic model. The earlier comparison, scalar study, and protected-observation experiment retain their original volatility parameters and their separate evidence records.

Training uses a global grid of 64 cells. Its reference policy is the cell-average analytical schedule and remains fixed through all three work levels. The cumulative occupation samples contain 256, 512, and 1,024 states. Each state comes from an independent reference trajectory at a uniformly selected decision node. Postdecision inputs are the pre-noise mean and the next decision date. NBO and Raw share four-antithetic continuation targets; all member rollouts and all reused labels are counted. These sample sizes define increasing information budgets, not equal floating-point work across methods.

At each stage NBO fits the spatial continuation derivative by full-replay Sobolev mean-square loss, with 1,200 Adam updates followed by L-BFGS. The latter has nominal iteration and evaluation limits of 300 and 400; the current line search is completed and every actual closure call is counted. A separate time component then receives 400 value-fitting updates with the spatial component fixed. NBO and Raw use the same feasible actor objective and minibatch/update streams, with 200, 400, and 800 additional actor updates at the three stages. DPO and HJB use 100, 200, and 400 additional updates. The minibatch size is 128; the actor, spatial-critic, time-component, and HJB learning rates are 0.003, 0.002, 0.003, and 0.002, respectively. Model weights and replay continue across stages. These are declared optimization procedures; a common sample budget does not make their optimization costs identical.

Online certification and independent final confirmation both use 2,048 decision cells and 8,192 paths per stream. The selected fixed networks are deployed on that finer grid. The learned 64-cell continuation derivative is consequently a predictor in the fine-grid mechanism assessment, whose error account includes that discrepancy; it is not identified with an exact fine-grid or continuous-time costate. The protocol and numerical source were frozen before the confirmation experiment. Exploratory pilots that selected the design remain distinct from the confirmatory observations.

The measured process clock begins before a fresh worker is launched and ends after its final evidence is durably written. It includes imports, model setup, de-

terministic constants, fitting, checks, weight reloading, confirmation, and output. Dependency installation and artifact extraction precede this clock and are not charged as method execution time.

9.2. Contribution-isolating comparisons

The comparison separates continuation regression, direct policy optimization, and direct approximation of the Bellman equation. The direct HJB method uses the same two learned scalar components as NBO: a time network of widths (16, 16) and a state–time network of widths (32, 32), each with hyperbolic tangent activations and a scalar output. Its known lift is the full-value lift, so the terminal condition is imposed exactly. NBO’s known lift instead represents its postdecision continuation target. This difference follows from the objects being approximated; the learned residual capacity is the same. Both methods use the prespecified occupation sample and the same available model information. No reference solution enters either training objective.

Write $c = a - (\sigma_I^2 + \sigma_C^2)/2$ and $\Sigma\Sigma' = \sigma_I^2 I + \sigma_C^2 \mathbf{1}\mathbf{1}'$. The direct method minimizes the squared residual of

$$\begin{aligned} 0 = & -v_t - \max_{m \in \mathcal{M}_t} \{u(y, m) + (c\mathbf{1} + \kappa \tanh(By) - m)' D_y v\} \\ & - \frac{1}{2} \text{tr}(\Sigma\Sigma' D_y^2 v) + \rho v, \quad v(T, y) = g(y), \end{aligned} \tag{44}$$

where $\mathcal{M}_t$ is the action tube shared by the four primary methods. For $q = dD_y v$, the action-dependent Hamiltonian is

$$H_q(m) = \frac{1}{d} \sum_i \log m_i - \frac{\zeta}{2} \bar{m}^2 - \frac{1}{d} \sum_i q_i m_i. \tag{45}$$

It is strictly concave. Its maximizer is obtained from the scalar consistency equation $z = d^{-1} \sum_i m_i(z)$, with $m_i(z) = \text{proj}_{[\ell_t, u_t]}(q_i + \zeta z)^{-1}$ when $q_i + \zeta z > 0$, and $m_i(z) = u_t$ otherwise. The implementation uses the same 44 monotone bisection steps as the other feasible costate comparisons. The direct HJB policy evaluates a critic gradient and this maximization at every decision. NBO deploys its separately learned actor. The resulting difference in online computation is included in validation, stopping, confirmation, and deployment costs.

Two independent Gaussian probe banks estimate the covariance Hessian trace. If $\hat{R}_1$ and $\hat{R}_2$ are the corresponding residual estimates, then, conditional on a collocation state and the current critic,

$$\mathbb{E}[\hat{R}_1 \hat{R}_2] = R[v]^2. \quad (46)$$

The product, rather than the square of one estimate, is the training loss. Its realized value can be negative. Negative batches are retained. This identity specifies an optimization objective; the common payoff verifier supplies the economic comparison.

Classical scalar training. The original volatility calibration also has a scalar comparison. A monotone implicit Howard method solves the one-state problem and saves an action at every time and state node. These saved policies are deployed by interpolation on fresh diffusion paths, together with the analytical schedule and the hash-pinned scalar NBO and DPO policies from the earlier experiment. Full-box and matched-tube classical policies are separate candidates. Three successively refined state–time grids are followed by an enlargement of the state interval at fixed spatial and temporal spacing. The deployment meshes aggregate the same fine Brownian increments. Policy differences are reported both over the declared initial population and at its individual states -1 , 0 , and 1 . Historical neural fitting clocks retain their historical interpretation; the new classical solves and all new deployment work are measured separately.

The scalar reference has a numerical error account at the level of the declared finite-grid equations. Its algebraic error account uses an outward residual calculation. State-domain approximation and diffusion time discretization remain separate from this algebraic account. The scalar comparison supplies an independent low-dimensional accuracy and work benchmark; the high-dimensional method comparison uses its own payoff data.

9.3. The distribution of a computational method

A computational method includes its initialization, training draws, validation rule, stopping rule, and response to failure. We specify these objects jointly. Let $\mathcal{S}$ be the sixteen complete execution streams listed in the protocol, fixed before

the confirmation experiment, and let $\pi_{m,s}$ denote the policy returned by method m under stream s . The randomized implementation draws s uniformly from $\mathcal{S}$; the experiment executes every member of this distribution. A failed fit returns the feasible analytical schedule and remains in the denominator. Conditional on the stated software environment, the stream specifies the complete training and checking computation. The distribution does not include arbitrary unrecorded initializations or a new draw of the stopping bank on re-execution.

Write $J(\pi)$ for the payoff in the original continuous-time economy, averaged over the specified nine initial-state profiles. The method-level objects are

$$G_m = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \{J(\pi_{m,s}) - J(\bar{\pi})\}, \quad (47)$$

$$\Delta_{m,b} = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \{J(\pi_{m,s}) - J(\pi_{b,s})\}. \quad (48)$$

The first object is improvement on the schedule. The second is the direct comparison of two randomized computational procedures with the same declared stream distribution. Enumeration accounts for the entire training distribution; there is no sampling error from an unobserved subset of these sixteen streams. The remaining uncertainty concerns their independent payoff simulations and the numerical transfer to continuous time. An assertion about this finite distribution is not an assertion about all possible training initializations.

The stopping target is a gain of 0.0005. At each of three declared work checkpoints, fitting pauses for a new, independent certification calculation. Training stops when its lower endpoint attains the target. Failed checks, initialization, action search, checkpoint input and output, and the independent final verification are included in the measured work. The gain 0.001 is a secondary final-attainment threshold; it is not a second measured stopping experiment. A failure to attain either target is retained.

9.4. Simultaneous inference for the finite method distribution

Each returned policy receives N fresh confirmation paths. The confirmation bank is independent of fitting, validation, and every stopping decision. Methods share

initial profiles and innovations within a stream, whereas the confirmation banks of distinct streams are independent. A direct contrast is formed on each common path. Its common analytical schedule cancels, and its transfer account contains the two policy allowances rather than two copies of the schedule allowance, when both entries are simulated candidate gains. An analytical fallback has the exact gain zero. A comparison with that fallback retains the other candidate's full schedule-relative transfer account; two analytical fallbacks have the exact difference zero.

For one registered endpoint, let X_{sj} be the numerical path statistic, $Y_{sj} = \max\{-C_s, \min(C_s, X_{sj})\}$, and let β_s enclose the difference between the continuous-time endpoint θ_s and $\mathbb{E}Y_{sj}$. Thus β_s includes the appropriate diffusion, implementation, clipping-tail, and arithmetic accounts. Put $C = \max_s C_s$, $n = |\mathcal{S}|N$, $\bar{\beta} = |\mathcal{S}|^{-1} \sum_s \beta_s$, and denote the pooled mean and unbiased sample variance by $\bar{Y}$ and $\hat{v}$.

PROPOSITION 10—Finite-distribution comparison: *Conditional on the complete executed policy ensemble, suppose the confirmation observations are independent across pairs (s, j) and the stated range and transfer accounts hold. For $0 < \delta < 1$, the interval*

$$\bar{Y} \pm \left\{ \sqrt{\frac{2\hat{v}\log(4/\delta)}{n}} + \frac{14C\log(4/\delta)}{3(n-1)} + \bar{\beta} \right\} \quad (49)$$

covers $|\mathcal{S}|^{-1} \sum_s \theta_s$ with probability at least $1 - \delta$. The observations need not be identically distributed across streams.

The confirmation family assigns 0.02 across 238 two-sided events: in each dimension, four schedule comparisons and three direct NBO comparisons, each reported for sixteen streams and their uniform method mean. Online stopping receives a separate 0.01 across its 384 possible checks. The mechanism and protected-observation families each receive 0.01. Their union therefore has error probability at most 0.05 for the new experiment. The historical experiment retains its separately stated confidence event. Unused checks do not release probability for additional comparisons.

The practical-equivalence margin is fixed at 10^{-4} payoff units. For a direct interval $[L, U]$, material superiority requires $L > 10^{-4}$, and practical equivalence requires $-10^{-4} < L \leq U < 10^{-4}$. Statistical superiority, $L > 0$, is reported separately. An

interval containing zero does not establish equivalence. Under log flow utility, the fixed margin corresponds to a proportional consumption fee $1 - \exp\{-10^{-4}/A\}$, where $A = (1 - e^{-\rho T})/\rho$; for the stated primitives this is approximately 1.0201 basis points.

Simultaneous stream intervals also enclose the probability that the randomized implementation attains a target g . If they are $[L_s, U_s]$, the corresponding bounds are

$$\frac{\#\{s : L_s \geq g\}}{|\mathcal{S}|} \leq \frac{\#\{s : \theta_s \geq g\}}{|\mathcal{S}|} \leq \frac{\#\{s : U_s \geq g\}}{|\mathcal{S}|}. \quad (50)$$

No binomial sampling interval is attached to a population that has been enumerated in full. Work is likewise reported over all streams, with unattained runs retained in the distribution of work to target. A median over successful runs alone is not the method's median.

The economic decomposition records production, consumption, and terminal dispersion contributions on the same numerical confirmation paths. These components add to the observed numerical payoff gain and help describe the policy's behavior. They have no separate continuous-time confidence statement: the transfer allowance is attached to the total payoff and is not divided among components without additional bounds.

9.5. Method performance, robustness, and complete work

The confirmation record contains all 128 declared method–dimension–stream executions. Tables II and III report the finite-distribution payoff objects defined in Section 9.3. Every endpoint is reconstructed from the original final arrays after checking their common-path identities and source manifests.

Tables IV and V measure the executed stopping procedures, including unsuccessful checks and final confirmation. Table VI describes the numerical channels of the payoff change without assigning the total continuous-time certificate to its separate components.

Economic performance of the declared methods. Continuation evaluation and feasible improvement produce positive continuous-economy gains in both dimensions. NBO's method means are approximately 0.001268 and 0.001342 in dimensions

TABLE II

PAYOFF OF THE FINITE-DISTRIBUTION COMPUTATIONAL METHODS

d	Method	Mean gain	Lower	Upper	At least target
10	NBO	0.001268	0.000943	0.001592	16/16
10	Raw	0.001267	0.000942	0.001592	16/16
10	Direct policy	0.001262	0.000938	0.001587	16/16
10	Neural HJB	0.000973	0.000647	0.001299	0/16
50	NBO	0.001342	0.000957	0.001727	16/16
50	Raw	0.001343	0.000958	0.001728	16/16
50	Direct policy	0.001347	0.000962	0.001732	16/16
50	Neural HJB	-0.000156	-0.000558	0.000246	0/16

Notes: Means average all sixteen declared execution streams. Endpoints enclose continuous-time payoff gains on the analytical schedule. The last column counts streams whose unrounded simultaneous final lower endpoint attains 0.0005; it is a lower bound on attainment under the declared finite distribution. No stream is excluded. Every lower endpoint is rounded down and every upper endpoint up to six decimal places; displayed intervals therefore contain the original protected intervals. Means are descriptive and rounded to nearest.

TABLE III

DIRECT METHOD COMPARISONS AT THE DECLARED ECONOMIC MARGIN

d	Comparison	Mean	Lower	Upper	Decision
10	NBO–Raw	0.000001	-0.000362	0.000364	Unresolved
10	NBO–Direct policy	0.000005	-0.000358	0.000368	Unresolved
10	NBO–Neural HJB	0.000295	-0.000073	0.000663	Unresolved
50	NBO–Raw	-0.000001	-0.000429	0.000426	Unresolved
50	NBO–Direct policy	-0.000005	-0.000432	0.000423	Unresolved
50	NBO–Neural HJB	0.001498	0.001051	0.001945	Superior

Notes: Pathwise differences use common profiles and innovations within each stream. The equivalence margin is 10^{-4} payoff units. Superior and inferior refer to economic differences exceeding that margin. Decisions use the original unrounded endpoints. An unresolved comparison is not evidence of equivalence. The schedule transfer cancels between two simulated candidates; comparison with an exact analytical fallback retains the other candidate's full schedule-relative transfer. Every lower endpoint is rounded down and every upper endpoint up to six decimal places; displayed intervals therefore contain the original protected intervals. Means are descriptive and rounded to nearest.

TABLE IV

COMPLETE MEASURED WORK OF THE STOPPING PROCEDURES

d	Method	Stopped	Mean (s)	Median (s)	Restricted (s)	MiB
10	NBO	0/16	134.82	–	900.00	474.3
10	Raw	0/16	122.67	–	900.00	473.0
10	Direct policy	0/16	285.53	–	900.00	489.4
10	Neural HJB	0/16	373.35	–	900.00	477.3
50	NBO	16/16	357.80	376.03	357.80	530.0
50	Raw	16/16	346.22	365.55	346.22	531.7
50	Direct policy	16/16	596.94	627.10	596.94	545.5
50	Neural HJB	0/16	1690.10	–	900.00	528.4

Stopped counts actual online certificates at the 0.0005 target. Mean is total consumed parent-clock time through independent final confirmation. Median is the lower 0.5 quantile of work to target, with failures assigned infinity; a dash denotes an unattained median. Restricted work is the mean of min(time to target, 900 seconds), retaining failures at 900. Memory is the maximum fresh-worker peak resident size over streams. These clocks describe the recorded execution environment.

TABLE V

RECORDED OPERATION COUNTS OF THE COMPLETE STOPPING PROCEDURES

d	Method	Train 10^6	Verify 10^6	Actor	Critic	Grad. 10^6	Search 10^6
10	NBO	0.20	134.22	1400	5700	1.04	0.00
10	Raw	0.20	134.22	1400	0	0.00	0.00
10	Direct policy	23.00	134.22	700	0	0.00	0.00
10	Neural HJB	0.07	134.22	0	700	67.20	2956.73
50	NBO	0.18	127.93	1250	5344	0.93	0.00
50	Raw	0.18	127.93	1250	0	0.00	0.00
50	Direct policy	23.00	134.22	700	0	0.00	0.00
50	Neural HJB	0.07	134.22	0	700	67.20	2956.73

Entries are means over all sixteen streams. Train and verify count simulator transitions; actor and critic count optimizer updates; grad. counts first-derivative rows; search counts action-search iterations, including HJB deployment. These are work proxies, not equivalent FLOPs. Full first- and second-derivative, forward, backward, and other counters are retained in the machine-readable report.

The parent clock additionally includes every failed operation and all I/O.

TABLE VI

DESCRIPTIVE DECOMPOSITION OF THE NUMERICAL PAYOFF GAIN

d	Method	Production	Consumption	Dispersion	Total
10	NBO	0.002303	-0.001271	0.000236	0.001268
10	Raw	0.002303	-0.001273	0.000237	0.001267
10	Direct policy	0.002279	-0.001248	0.000231	0.001262
10	Neural HJB	0.001830	-0.001009	0.000152	0.000973
50	NBO	0.002297	-0.001343	0.000387	0.001342
50	Raw	0.002297	-0.001342	0.000389	0.001343
50	Direct policy	0.002275	-0.001310	0.000382	0.001347
50	Neural HJB	0.000910	-0.001517	0.000451	-0.000156

Entries are observed finite-step common-path contributions, averaged uniformly over the complete stream set. Consumption is minus the consumption deficit; dispersion is the terminal contribution. Their pathwise sum equals the numerical gain. No component continuous-time confidence interval or causal attribution is asserted; the separate total-payoff account remains authoritative.

ten and fifty; their simultaneous lower endpoints exceed 0.000943 and 0.000957, respectively. These statements average over all sixteen declared complete algorithm streams. All 128 executions returned fitted policies; none used the analytical fallback.

The direct comparison identifies a material advantage over the specified neural HJB procedure in dimension fifty. With the original transfer account, the NBO-minus-HJB interval is contained in $[0.001051, 0.001945]$, whose lower endpoint exceeds the prespecified 10^{-4} economic margin. The original ten-dimensional interval, contained in $[-0.000073, 0.000663]$, is inconclusive. All four direct comparisons with cached Raw and direct policy optimization are also inconclusive under that account. Thus the experiment establishes a domain in which the NBO evaluation-actor construction improves economic performance relative to direct residual fitting with the stated common representation capacity. The finding concerns the declared architectures, training schedules, stopping rule, and finite stream distribution; it is not a ranking of every possible implementation of the three alternatives.

Economic accuracy and complete work. In dimension fifty, all sixteen NBO, Raw, and DPO executions attained the online gain target 0.0005, while no HJB execution

1 did. Their mean complete work was approximately 357.80, 346.22, 596.94, and 1
 2 1690.10 seconds, respectively, including unsuccessful checks and final confirmation. 2
 3 NBO therefore attained the same declared economic target with less measured 3
 4 work than DPO on the recorded environment; Raw was cheaper than NBO. HJB's 4
 5 unattained runs remain in the work distribution. The reported 900-second restricted- 5
 6 mean functional for an unattained target is a censoring convention, not the actual 6
 7 work consumed or an assertion that the run stopped at 900 seconds. 7

8 No ten-dimensional execution attained the online stopping target. Independent 8
 9 final confirmation nevertheless places every returned NBO, Raw, and DPO policy 9
 10 above 0.0005 in that dimension, as it does in dimension fifty. These final certificates 10
 11 do not alter the earlier stopping decisions. No method has a positively certified 11
 12 stream attaining the secondary 0.001 threshold. The complete stream and check- 12
 13 point tables retain this difference between measured stopping and final economic 13
 14 accuracy. 14

15 *Sources of the observed gain.* The numerical decomposition records how the 15
 16 learned decisions change the capital economy. For NBO, production contributes 16
 17 approximately 0.002303 and 0.002297 in dimensions ten and fifty, while the con- 17
 18 sumption contribution is approximately -0.001271 and -0.001343 . The terminal- 18
 19 dispersion contribution is approximately 0.000236 and 0.000387. Their sums equal 19
 20 the observed numerical gains. These quantities describe the production benefit 20
 21 and the consumption cost of state-dependent decisions on the same paths; the 21
 22 continuous-time confidence statement belongs to the total payoff, not separately to 22
 23 these components. 23
 24 24

25 9.6. Direct comparisons with a shared diffusion-transfer account 25

26 26
 27 The direct statistic already compares policies on common innovations. The de- 27
 28 terministic transfer can use that same coupling. Proposition S.6 in the supplement 28
 29 bounds the error of the payoff difference for two held policies directly, retaining 29
 30 the original decision grid and realized actions. The fitted policies, simulated obser- 30
 31 vations, stopping decisions, clipping, tail account, and simultaneous confidence 31
 32 family remain unchanged. 32

The following table reports the original and refined endpoints together. The refinement concerns a proved deterministic allowance; it is not an additional simulation experiment or a retrospectively changed stopping rule. Comparisons involving a schedule fallback retain their stated fallback account. The refined comparison

TABLE VII
ORIGINAL AND REFINED DIRECT METHOD ENDPOINTS ON THE SAME OBSERVATIONS

d	Comparison	Mean	Original interval	Refined interval	Decision
10	NBO–Raw	0.000001	[-0.000362, 0.000364]	[-0.000110, 0.000112]	Unresolved
10	NBO–Direct policy	0.000005	[-0.000358, 0.000368]	[-0.000106, 0.000117]	Unresolved
10	NBO–Neural HJB	0.000295	[-0.000073, 0.000663]	[0.000179, 0.000411]	Superior
50	NBO–Raw	-0.000001	[-0.000429, 0.000426]	[-0.000133, 0.000130]	Unresolved
50	NBO–Direct policy	-0.000005	[-0.000432, 0.000423]	[-0.000137, 0.000127]	Unresolved
50	NBO–Neural HJB	0.001498	[0.001051, 0.001945]	[0.001347, 0.001650]	Superior

The new deterministic common-innovation transfer is proved in Appendix E. Both intervals use the identical original clipped statistic, path identities, empirical Bernstein event, and confidence allocation. The refinement was developed after the original source freeze and inspection of the first transfer width.

Only the deterministic transfer changes; mixed analytical fallbacks retain the original allowance. Decisions use the original 10^{-4} economic margin. Original training, stopping, and work records remain unchanged.

establishes economically material NBO gains over the specified neural HJB method in both dimensions. Its direct lower endpoints exceed 0.000179 and 0.001347, respectively, and hence exceed the original 10^{-4} margin. The ten-dimensional comparison changes from unresolved under the original account to materially positive under the proved refinement; the fifty-dimensional conclusion is strengthened. The four NBO–Raw/DPO intervals remain unresolved: they contain zero and extend outside the practical-equivalence band. They also do not certify noninferiority at the stated margin. The conclusion follows from a sharper deterministic account of the same paired observations, with the original statistical and computational records retained.

9.7. Absolute accuracy in the original control problem

Let ν be the declared uniform distribution over the nine initial profiles, and let J_ν^0 denote the analytical schedule's payoff. The pointwise anchor account in (S.14) implies $V_\nu^* - J_\nu^0 \leq B_{d,\nu}$ after averaging over this initial population. The repeated worker records concern this same bound. We verify their profiles and component weights and retain the maximum of the already averaged upper bounds if their final rounding differs.

For a method m , write $\bar{J}_{d,m} = 16^{-1} \sum_s J_\nu(\pi_{m,s})$. On the simultaneous confirmation event, its existing gain lower endpoint satisfies $\bar{J}_{d,m} - J_\nu^0 \geq L_{d,m}$. Consequently,

$$0 \leq V_\nu^* - \bar{J}_{d,m} \leq B_{d,\nu} - L_{d,m}. \quad (51)$$

The optimum in this expression ranges over the original adapted action-box class. The returned methods use the declared common tube, which is a feasible subset. The inequality therefore accounts for absolute loss against the original economic problem. It does not identify the unobserved loss or require another probability allocation.

The reported full-class upper bounds range from 0.025971 to 0.038358 payoff units. Their ratios to the declared margin range from 259.8 to 383.6. Thus the current absolute account gives guarantees at those explicit tolerances. It does not certify proximity to the full optimum within the 10^{-4} margin, even when a relative method comparison is resolved.

Table VIII puts this absolute account next to the economic margin used for direct method comparisons. A positive gain lower endpoint establishes improvement on the schedule; a direct comparison resolves a difference between two trained methods; and a small right-hand side of (51) would establish proximity to the full optimum. These implications concern different quantities. The neural HJB comparator supplies an independent candidate construction, while the scalar calculation supplies a separate finite-grid accuracy account. Neither supplies an exact high-dimensional reference value.

TABLE VIII

FULL-CLASS PAYOFF-GAP UPPER BOUNDS AND THE ECONOMIC MARGIN

d	Method	Gain lower	Regret upper	Margin	Ratio
10	NBO	0.000943	0.025971	0.0001	259.8
10	Raw	0.000942	0.025971	0.0001	259.8
10	Direct policy	0.000938	0.025976	0.0001	259.8
10	Neural HJB	0.000647	0.026266	0.0001	262.7
50	NBO	0.000957	0.036843	0.0001	368.5
50	Raw	0.000958	0.036841	0.0001	368.5
50	Direct policy	0.000962	0.036838	0.0001	368.4
50	Neural HJB	-0.000558	0.038358	0.0001	383.6

The regret upper bound is $B_{d,v} - L_{d,m}$, rounded outward: $B_{d,v}$ bounds the analytical schedule's loss against the original full adapted control class, and $L_{d,m}$ is the existing lower endpoint for the uniform mean over all sixteen returned policies. The ratio divides the regret upper bound by the prespecified 10^{-4} payoff margin. All 128 source-bound confirmation records are audited. This deterministic translation uses the existing confirmation event and no additional alpha.

9.8. Continuation accuracy where the policy changes the allocation

The mechanism assessment follows the policy that each registered NBO stream actually returns. It draws fresh occupation paths and uniform action bridges after all policy selection. The reference remains the cell-average analytical schedule on one global assessment grid. The saved scalar critic may have been fitted on a coarser grid; its assessment error includes this discrepancy.

Table IX compares the fitted costate with a direct four-path antithetic bank on the identical continuation queries. The calculation uses independent cross-products to remove reference-bank variance from the critic's risk estimand. It also reports the paired difference in risk, retaining negative sample contributions and all registered streams. In dimension 10, the costate-risk contrast has endpoints -0.0344447 and 0.0343934; this interval does not determine which predictor has smaller risk.

In dimension 50, the costate-risk contrast has endpoints -0.0225684 and 0.0225961; this interval does not determine which predictor has smaller risk.

Table X carries the same continuation account through the feasible actor’s actual action change and the quantitative continuous-payoff transfer of Theorem 9. The approximate continuation’s scalar advantage is evaluated at both action endpoints. The actor’s gap is bounded from the saved global curvature certificate, including the endpoint calculation when strong concavity is unavailable. The combined Bellman-bridge lower endpoint in dimension 10 is -0.006778. The retained costate, sampling, and implementation allowances do not establish improvement through this sufficient bound.

The combined Bellman-bridge lower endpoint in dimension 50 is -0.004800. The retained costate, sampling, and implementation allowances do not establish improvement through this sufficient bound.

The complete six-statistic intervals and numerical cushions appear in Table K.I. Table K.II reports the additional fresh-process assessment work. These calculations distinguish accuracy of the evaluation stage, quality of the feasible action update, and the independent payoff comparison between complete algorithms.

TABLE IX

COSTATE PREDICTION ON THE CANDIDATE’S OCCUPATION LAW

d	Critic risk mean	Raw risk mean	Contrast lower	Contrast upper
10	0.045	0.302	-344.447	343.934
50	0.398	0.259	-225.684	225.961

Notes: All displayed risk entries are multiplied by 10^4 . Each dimension pools 256 independent action bridges from every one of the 16 registered NBO streams, or 4,096 bridges. Each bridge has two independent four-path antithetic continuation banks on the 2,048-cell global grid. The signed critic statistic removes reference Monte Carlo variance in expectation. The raw statistic is the risk of one four-path bank on the identical represented queries. The paired contrast estimates critic risk minus raw risk. Its endpoints use the registered family of 12 two-sided statements with total error probability 0.01 and include interval sample arithmetic and proved clipping tails. These are evaluation-subproblem comparisons; the returned Raw policy’s welfare is assessed separately.

TABLE X

FROM THE CONTINUATION ACCOUNT TO ECONOMIC GAIN

d	L_M	$\sqrt{U_A U_E}$	U_D	Holding/transfer	Gain lower
10	-0.000849	0.005630	0.004113	0.000300	-0.006778
50	-0.000153	0.004285	0.002962	0.000363	-0.004800

Notes: The costate upper bound includes the exact-state representation allowance. The holding and transfer column sums the outward holding deficit and the quantitative paired payoff transfer to the original continuous economy. The last column is the larger of the product bound and the optional quadratic actor-gap bound when its positive modulus is verified. A negative endpoint records the limit of this sufficient mechanism account. Actual method welfare remains assessed by the independent direct payoff experiment; the mechanism assessment cannot change selection or stopping.

9.9. Independent scalar training and numerical accuracy

The separate scalar study uses the original volatility calibration, $(\sigma_I, \sigma_C) = (0.15, 0.10)$, and the same one-state capital economy. Eight independent Howard solves generate full-box and matched-tube policies on all four prespecified grids. The policies are then evaluated by actual interpolation-based deployment on 8,192 fresh paths at both 1,024 and 2,048 time steps. The analytical schedule and both frozen scalar neural policies use the same initial states and Brownian paths.

All 8 classical solves reached the stated stopping criterion, using at most 3 Howard updates at each date. Their finite-grid algebraic bounds range from 1.072×10^{-12} to 8.938×10^{-12} . The last joint state–time refinement changes the three initial-state values by amounts between 1.724×10^{-4} and 3.250×10^{-4} . The subsequent enlargement of the domain changes these values by at most 4.441×10^{-16} . Thus the algebraic solve, the training-grid approximation, and the finite-domain approximation have separate, observable numerical accounts. The small domain change does not bound the remaining time or spatial discretization error.

At 2,048 deployment steps, the largest-domain classical policy has a mean schedule gain of 7.819×10^{-4} , compared with 7.412×10^{-4} for NBO and 7.065×10^{-4} for DPO. The paired classical advantages are 4.069×10^{-5} (Monte Carlo standard error 6.841×10^{-7}) against NBO and 7.541×10^{-5} (4.063×10^{-7}) against DPO. Both advantages are positive at each of the three reported initial states. The numerical

TABLE XI
INDEPENDENT SCALAR TRAINING AND ALGEBRAIC ACCURACY

Policy	$N_y/N_t/L$	$v(-1)$	$v(0)$	$v(1)$	E_0	Seconds
F1	401/256/6	-3.393994	-1.314658	0.816406	1.08E-12	3.76
F2	801/512/6	-3.393648	-1.314316	0.817061	2.28E-12	9.30
F3	1601/1024/6	-3.393474	-1.314143	0.817386	5.93E-12	24.54
F4	2401/1024/9	-3.393474	-1.314143	0.817386	8.94E-12	30.26
T1	401/256/6	-3.393994	-1.314658	0.816406	1.08E-12	3.77
T2	801/512/6	-3.393648	-1.314316	0.817061	2.28E-12	9.33
T3	1601/1024/6	-3.393474	-1.314143	0.817386	5.93E-12	24.64
T4	2401/1024/9	-3.393474	-1.314143	0.817386	8.94E-12	30.50

F1–F4 solve the full primitive action box; T1–T4 solve the common schedule tube of radius 0.1. E_0 is the outward finite-grid algebraic error bound in Proposition S.15. The displayed error endpoints are rounded upward; the full outward endpoints are retained in the artifact. Seconds include each new solve, its interval residual audit, and policy serialization. The last grid enlarges the domain from $[-6, 6]$ to $[-9, 9]$ at the same declared spatial and temporal spacing. Boundary data are the stated far-state approximation.

comparison therefore identifies an independently computed feasible classical competitor with higher observed payoffs than both frozen neural policies in this scalar economy.

The neural ordering varies across initial states. NBO minus DPO has population mean 3.472×10^{-5} , but its conditional mean at $y = 1$ is -6.206×10^{-5} . The same reversal occurs on the 1,024-step deployment mesh. This state-specific outcome is retained alongside the population comparison. Corresponding full-box and tube policies have identical payoffs path by path on both reported deployment meshes. The maximum recorded nodal deviation from the schedule is 0.046600, below the tube radius. This observation concerns the computed scalar policies and does not establish that the tube restriction is harmless in every dimension.

The complete full-box and matched-tube value arrays coincide bit for bit on each of the four grids. Because the action sets are nested and the terminal and boundary data are identical, Corollary S.16 converts the two algebraic enclosures into a bound on the loss from the tube restriction in the exact finite-grid problem.

TABLE XII

DEPLOYED SCALAR POLICY GAINS ON COMMON DIFFUSION PATHS

Policy	$y = -1$	$y = 0$	$y = 1$	Population	MC s.e.
Schedule	0.0000	0.0000	0.0000	0.0000	0.0000
F1	0.1137	1.5586	0.6768	0.7818	0.0069
F2	0.1137	1.5587	0.6771	0.7819	0.0069
F3	0.1136	1.5586	0.6772	0.7819	0.0069
F4	0.1136	1.5586	0.6772	0.7819	0.0069
T1	0.1137	1.5586	0.6768	0.7818	0.0069
T2	0.1137	1.5587	0.6771	0.7819	0.0069
T3	0.1136	1.5586	0.6772	0.7819	0.0069
T4	0.1136	1.5586	0.6772	0.7819	0.0069
NBO (R12)	0.1036	1.5557	0.5683	0.7412	0.0071
DPO (R12)	0.0468	1.4460	0.6303	0.7065	0.0066

All payoff gains and standard errors are multiplied by 10^3 . The comparator is the analytical schedule evaluated on the same numerical mesh and Brownian path. Population denotes the mean over independent uniform initial-state draws; the last column is its paired Monte Carlo standard error. Initial-state columns retain their separate rankings. NBO and DPO are the prespecified frozen R12 scalar policies. These are conditional Euler-deployment diagnostics; Monte Carlo standard errors do not include continuous-time or boundary approximation error.

Across the four grids, this loss at every initial-date node is at most 1.788×10^{-11} . This deterministic conclusion concerns the stated finite-grid problem; the domain and diffusion approximation accounts retain their separate scope.

The complete scalar process consumed 195.90 seconds, including fresh-process setup, all eight solves, interval audits, serialization, both common-path deployments, and output. The individual new classical solves consumed 3.76–30.50 seconds. Within the fine-mesh deployment, the measured action-evaluation totals were 1.26–1.36 seconds for the classical tables and 9.38–9.60 seconds for the frozen neural maps. These are operation-specific measurements on the recorded worker. The historical neural fitting clocks are not included in a new end-to-end training comparison.

The algebraic error bound concerns the exact equations with their declared boundary data. The independent domain enlargement and paired deployment-mesh comparisons quantify distinct numerical sensitivities. The complete tables retain all trained policies, both meshes, all initial-state strata, and all 110 pairwise numerical comparisons. They supply a scalar accuracy and classical-work benchmark without identifying an exact value for the high-dimensional diffusion.

9.10. *Implementation from finite state measurements*

The implementation question has a direct economic interpretation. A manager observes a vector of log capital stocks at specified dates, computes consumption rates with the saved network, and holds those rates until the next observation. The manager's input consists of the measured physical state. The finite-observation controller does not require a Brownian coordinate or the intervening drift integral. Proposition S.13 applies to this controller through a protected parent with the same saved weights and inward action guard.

The experiment verifies that parent afresh at 1,024 and 4,096 decision dates in dimensions 10 and 50. It retains the original volatility calibration and the previously fixed iteration-80 NBO weights; no outcome in the new observation experiment selects a policy. Table XIII subtracts each deterministic sensing allowance from the corresponding parent guarantee. All four parents have positive lower endpoints. Six of the twelve finite-observation lower endpoints remain positive. With 4,096 dates and exact measurements, the continuous-economy gain is at least 0.001050 in dimension 10 and 0.001054 in dimension 50. In dimension 10, a measurement error of 0.0001 retains a lower gain of 0.000889 at the finer grid.

The dependence on measurement precision is economically material. At dimension 50, an error of 0.0001 already exceeds the available parent margin under the current global network bound, and an error of 0.001 removes a positive guarantee in both dimensions. These endpoints give an implementation requirement for the specific measured-state rule. They do not infer the sign of its actual gain when the allowance is too large. The full intervals, allowance-to-gain ratios, and fresh-process costs appear in Tables M.V and M.VI.

TABLE XIII
CONTINUOUS-ECONOMY GAIN BOUNDS WITH FINITE OBSERVATIONS

d	N	Parent lower	$\nu = 0$	$\nu = 0.0001$	$\nu = 0.001$
10	1024	0.000766	0.000687	0.000526	-0.000925
10	4096	0.001070	0.001050	0.000889	-0.000559
50	1024	0.000807	0.000252	-0.001038	-0.012651
50	4096	0.001193	0.001054	-0.000239	-0.011873

Notes: All lower endpoints are printed downward to six decimal places. The last three columns are lower bounds for the finite-observation controller’s payoff gain over the analytical schedule. Each parent is freshly verified at the same decision grid as its sensor transfer, using 8,192 independent paths. The four parent intervals share error probability 0.01; the transfers are deterministic. The fixed weights and original volatility calibration are retained from the reviewed study. A negative endpoint means that this error account does not certify improvement at that observation precision.

10. CONCLUSION

Neural Bellman Operators organize economic computation around the continuation object that is evaluated, the feasible action that it changes, and the economic consequence of that change. The separation matters for recursive utility, preference adjustment, temporal selves, and strategic interaction because each problem has its own evaluation equation and deviation criterion. The approximation and verification arguments make those distinctions explicit.

The capital economy permits an exact finite Bellman account under the candidate’s occupation law, a computable feasible-actor gap, and a quantified transfer to the original continuous diffusion. These results make the evaluation–action–payoff chain executable without identifying a finite-rollout derivative with a continuous-time costate. The independent paired transfer also makes use of the actual common innovations in direct method comparisons, while preserving the original observations and confidence event.

The complete sixteen-stream experiment establishes economically material NBO payoff gains over the declared neural HJB implementation in both ten and fifty dimensions. In dimension fifty, NBO reaches the prescribed online gain target in every stream with less measured complete work than direct policy optimization. Cached Raw remains cheaper; the direct NBO–Raw and NBO–DPO payoff intervals

do not resolve their ranking or establish practical equivalence. Thus the experiment identifies a useful domain for the NBO evaluation–actor construction under an explicit computational contract.

The separate mechanism calculation reports all occupation, action, sampling, and numerical allowances, but its sufficient lower bound does not certify positive mechanism welfare at the executed precision. The full-class regret bounds likewise remain wider than the direct-comparison margin. The independent scalar calculation provides a sharper finite-grid account with a classical training and deployment comparison, and preserves the observed state-dependent ranking reversal. These qualifications attach each accuracy statement to the economic and numerical object that it actually controls.

Finite observations also support an operational use of the computed policy. A manager receives measured capital and selects consumption at stated dates; after the allowance for that grid and measurement precision, six of twelve specifications retain a positive continuous-economy lower gain. The original recursive-utility, endogenous-preference, temporal-self, and game applications retain their complete models and proofs. Together they state the economic conditions under which an evaluated continuation object can guide feasible decisions, while the capital experiments quantify the resulting payoff and computational tradeoffs.

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